Salary Increases and Skills-Based Bonuses: Teacher Retention and Workforce Composition

Adriano De Falco
Lund University

October 2026

Abstract

This paper studies the combined effect of unconditional salary increases and performance-based bonuses on teacher retention and workforce composition. I exploit Chile's *Carrera Docente*, a reform that raised base salaries by approximately 20% for public school teachers and introduced a career stage bonus linked to evaluation scores. Because the reform initially applied only to public schools, I use a difference-in-differences design and find that exit rates fell by approximately 18%. The effect, however, is concentrated among low-value-added teachers. A regression discontinuity design shows that the performance-based bonus has no detectable additional retention effect beyond the base salary increase.

Keywords: teacher retention, teacher quality, salary increases, performance-based bonuses

JEL codes: I21, J31, J45, M52

I am grateful for the valuable comments by Thomas Crossley, Jonathan Goupille-Lebret, Andrea Ichino, Vincenzo Scrutinio, Desislava Tartova and Giulio Zanella as well as participants to the EUI Econometrics Working Group, the 3rd ENS de Lyon – University of Bologna Workshop and the CEN Workshop of the Copenhagen Education Network. I thank the Departamento de Evaluación, Medición y Registro Educacional (DEMRE) for providing data on teachers' careers. I thank the Agencia de Calidad de la Educación for the student-level data.

1 Introduction

Teacher shortages are a growing concern across both developed and developing countries (UNESCO, 2023). In the United States, approximately 8% of public school teachers leave the profession each year (National Center for Education Statistics, 2023), with comparable or higher rates in other OECD countries (OECD, 2025). Moreover, several studies show that teacher turnover disrupts student learning, particularly in disadvantaged schools (Ronfeldt et al., 2013), and replacing teachers is expensive for school systems that must recruit and train new staff (Ingersoll, 2001). At the same time, a large body of evidence shows that teachers vary substantially in their effectiveness (Rivkin et al., 2005; Chetty et al., 2014; Jackson, 2018), making it important not only to retain teachers in general but to understand which teachers are retained.

Raising teacher compensation is a natural response, but its effect on the composition of the workforce depends on how the raise is designed. If part of the raise is tied to evaluated performance, this creates a wage gradient that favors better teachers, provided that evaluations capture actual teaching effectiveness. Combining a uniform raise with a performance-based component is therefore appealing, as the raise has the potential to broadly retain teachers, while the bonus differentially rewards the more effective ones. This combination, however, may not necessarily imply that better teachers are more likely to stay in the profession. In a standard Roy model (Roy, 1951), the extent to which a raise reduces exit is proportional to the utility it buys and the teacher's proximity to the exit margin. Assuming utility over income is concave, the base salary increase may reduce both, decreasing the retention value of the performance-based bonus. The retention response then concentrates among teachers who are near-indifferent between staying and leaving, and whether these marginal teachers are of high or low ability is an empirical question.

I study this using a natural experiment provided by Chile's *Carrera Docente*, a career ladder introduced in 2016 that restructured the compensation of public school teachers around a five-tier system.[1] The reform introduced an unconditional baseline salary increase of approximately 20% for all public school teachers, alongside a career stage

[1]Several countries have adopted similar career ladder systems. The closest comparators to the Chilean reform are Colombia's merit-based teacher career system (Busso et al., 2024), Peru's *Carrera Pública Magisterial* (Tournier and Chimier, 2019), and Mexico's *Carrera Magisterial* (Santibáñez et al., 2007). In the United States, Tennessee operated a statewide career ladder from 1984 to 1997 (Dee and Keys, 2004), and Washington D.C.'s IMPACT program links teacher pay and career progression to evaluation scores (Dee and Wyckoff, 2015). Singapore operates a multi-track career ladder in which advancement is based on principal-led qualitative assessments rather than standardized instruments.

bonus that depends on results in two mandatory evaluations (a teaching portfolio and a subject-knowledge exam). Moreover, it was initially applied only to public schools, which at the time employed roughly half of Chile's teachers; subsidized (voucher) schools became eligible only years later. Voucher school teachers thus provide a natural comparison group for the overall effect of the reform, that is, the combination of the baseline raise and the performance bonus. To isolate the effect of the bonus, I exploit the cutoff scores on the two mandatory evaluations that determine advancement across career stages.

Using administrative records covering the universe of primary and secondary school teachers in Chile over the period 2012–2018 (approximately 200,000 unique teachers and 830,000 teacher-year observations), I estimate the overall effect of the salary increase on teacher exit in a difference-in-differences design that compares public school teachers to their counterparts in subsidized (voucher) schools. The event study estimates show no differential trend before the reform and a growing reduction in exit rates afterward. By 2018, public school teachers are 0.7 percentage points less likely to exit relative to the pre-reform baseline, a decline of approximately 18%. Using imputed wages from the statutory salary schedule, this implies an elasticity of exit with respect to wages of approximately -0.8.

To understand which teachers are more likely to be retained, I develop a simple model of teacher exit decisions. A teacher remains in the profession as long as the utility she derives from teaching, which depends on income and on her intrinsic motivation for the job, exceeds the value of her outside option. Utility over income is concave, and both intrinsic motivation and outside options may vary with ability, which evaluation scores measure with noise. In this setting, a uniform raise retains the teachers who were closest to indifference between staying and leaving. Whether these marginal teachers are of high or low ability depends on how the two sources of heterogeneity vary with ability. If outside options improve with ability, effective teachers move closer to the exit margin and the raise is more likely to retain them. If instead effective teachers derive greater intrinsic satisfaction from teaching, they are pushed away from the exit margin and the raise is more likely to retain the less effective ones. A bonus tied to evaluation scores favors high-ability teachers, who are more likely to receive it. However, because the bonus is paid on top of the base raise, the utility it adds is evaluated where marginal utility is lower.

To measure ability empirically, I estimate teacher value-added (TVA) from standard-

ized test scores in mathematics and reading, linking teachers to the universe of their students. The estimation follows a standard value-added model in which current test scores are a function of lagged scores, student and classroom characteristics (class size, gender composition, parental education and income). I use each teacher's most recent pre-reform TVA as a fixed measure of ability, which is available for approximately 15% of the pre-reform analysis sample, corresponding to about 19,000 teachers. Within this subsample, I estimate a negative effect on exit that is however concentrated among low-TVA teachers. Their exit rate decreases by 1.10 percentage points, while the effect for high-TVA teachers is indistinguishable from zero. Two pieces of evidence help interpret this pattern in light of the theoretical framework. On the one hand, TVA is positively correlated with evaluation scores (a one-standard-deviation increase in the portfolio or subject-knowledge exam score is associated with increases of 0.03 and 0.02 in TVA, respectively), confirming that high-ability teachers can expect larger career-stage bonuses. On the other hand, low-ability teachers have higher pre-reform exit rates, placing them closer to the exit margin. The higher expected bonus therefore does not make high-ability teachers more responsive to the reform. A remaining question is whether this is despite the bonus having some retention effect, or whether the bonus has no effect at all.

I test this directly using a regression discontinuity design. Under the career system, teachers advance through stages based on whether their evaluation scores cross predetermined thresholds. Evaluations are mandated by law every four years, so that roughly one quarter of the workforce is assessed each year, and the year in which each teacher is evaluated is assigned by the education authority rather than chosen by the teacher. Evaluated teachers therefore do not self-select, and I show that the evaluated sample is comparable to the overall population in observables. Not all stage transitions are suitable for analysis. Higher stages are difficult to reach, so few teachers score near the corresponding thresholds, and the third stage is where most teachers will remain for most of their careers. I therefore focus on the transition from the second to the third stage, which carries an approximately 11% salary increase. Career stage assignment depends on two evaluation scores, the teaching portfolio and the subject-knowledge exam. I use the portfolio score as the running variable, recentered at the threshold that applies conditional on the subject-knowledge score. Using approximately 20,000 teachers evaluated in 2016 and 2017, I find no detectable evidence that the bonus reduces exit, with a point estimate close to zero. I also estimate how the effect of crossing the portfolio

threshold varies with the subject-knowledge score, and find that the effect remains close to zero across the support of the subject-knowledge score. This exercise extends the estimates beyond a single cutoff and shows that the result holds for a broader range of teachers.

Related literature. This paper contributes to several strands of the literature. First, it speaks to the large body of work on teacher compensation and retention. Many papers find that higher salaries reduce teacher turnover, though the magnitude of the response varies across contexts, sources of pay increases, and identification strategies. Elasticities are systematically higher when bonuses target teachers in hard-to-retain schools, in the range of -3 to -8 (Clotfelter et al., 2008; Springer et al., 2016; Tartova, 2025; Cowan and Goldhaber, 2018). When estimates exploit variation in wages or outside options, capturing a more representative group of teachers, the implied elasticities are lower, in the range of -1 to -1.5 (Falch, 2011; Hendricks, 2014), and closer to the estimate I obtain. I contribute to this literature by studying a national reform that restructured compensation for the entire public teaching workforce, providing a retention elasticity estimate that is directly relevant for the design of large-scale salary policies. Unlike studies that exploit targeted bonuses or local wage variation, my estimates capture the retention response of the entire public teaching workforce to a nationwide change in the salary schedule.[2]

Second, this paper relates to the literature on incentive contracts for public sector workers, and teachers in particular (Finan et al., 2017). A growing body of work studies the design of teacher performance pay, distinguishing between output-based incentives tied to student test scores (Lavy, 2009; Duflo et al., 2012; Muralidharan and Sundararaman, 2011; Leaver et al., 2021) and input-based incentives that reward teaching skills (Taylor and Tyler, 2012; Araya-Córdova et al., 2025). A key motivation for the latter is that tying pay to test scores may distort effort away from less measurable but equally valuable dimensions of teaching (Holmstrom and Milgrom, 1991; Glewwe et al., 2010; Dinerstein and Opper, 2022). I study the retention effects of an input-based contract with a base salary increase and an input-based bonus, where inputs are positively correlated with teacher value-added. I show that the input-based bonus

[2] A complementary literature studies how labor market conditions affect the quality and flow of teacher entrants rather than the retention of incumbents (Bacolod, 2007; Deneault, 2025; Fraenkel, 2022; Nagler et al., 2020).

has no additional effect on retention beyond the base salary increase.[3][4] This suggests that while performance-based evaluations may incentivize effort and professional development (Taylor and Tyler, 2012; Araya-Córdova et al., 2025), one should not expect performance-based bonuses to deliver additional retention benefits when accompanied by a substantial unconditional raise.

Third, this paper contributes to the literature on the interaction between wage compensation and teaching quality. A large body of work shows that performance pay increases teacher effort (Lavy, 2009; Duflo et al., 2012; Muralidharan and Sundararaman, 2011; Araya-Córdova et al., 2025) and can attract and retain more effective teachers (Biasi, 2021; Dee and Wyckoff, 2015; Johnston, 2025). Since the reform combines a baseline salary increase with a performance-based bonus,[5] and I show that the bonus has no additional retention effect, the paper also relates more closely to the literature on the impact of unconditional wage increases on student outcomes (De Ree et al., 2018; Bobba et al., 2021), where the evidence suggests that raising pay uniformly does not by itself improve performance. I contribute to this literature by focusing on the retention margin. I ask whether the composition of the teaching workforce, in terms of quality, changes as compensation increases.[6] I find that the reform disproportionately retains low-value-added teachers. This contrasts with Tartova (2025), who studies a 2014 reform that raised the bonus paid to teachers in France's most disadvantaged middle schools and finds that the retention effect is concentrated among high-value-added teachers.[7]

Within the same reform, Araya-Córdova et al. (2025) study a different margin. Exploiting the timing of the evaluations, they isolate the effect of being evaluated from that of the unconditional salary increase, and find that being evaluated raises effort and student achievement only for the small share of teachers facing strong incentives, with

[3] Indirect evidence consistent with nonlinear effects of salary increases includes the declining turnover elasticity with experience documented by Hendricks (2014) and the similar retention effects of bonuses of very different sizes across contexts (Springer et al., 2016; Clotfelter et al., 2008).

[4] Berlinski and Ramos (2020) also use a regression discontinuity design to study the retention effect of a skill-based bonus in Chile (the AEP award), finding no effect. Their setting differs from mine in that the AEP targets top-performing teachers, a selected population for whom the additional wage premium may have limited marginal value.

[5] The evaluations used for career advancement are positively correlated with teacher value-added, connecting this paper to the literature on the determinants of teaching effectiveness and whether evaluation instruments capture it (Rockoff et al., 2011; Chetty et al., 2014; Bacher-Hicks, 2022).

[6] An unconditional salary increase may also affect the quality of teacher entrants. See Dal Bó et al. (2013) and Nagler et al. (2020) for evidence on how compensation levels shape the quality of candidates entering public sector positions.

[7] Bates and Johnston (2025) also find no differential retention by quality, though their variation comes from pension changes, which primarily affect teachers near retirement age, while mine comes from salary changes that shift the incentive to stay for teachers across the entire career.

no effect on average. I look instead at the selection margin. I isolate the contribution of the two components of the reform to retention, and ask whether the teachers retained are of high or low ability, and find that they are of low ability. Together with the finding that incentives have no effect on average, this suggests that raising teacher pay may involve a trade-off between retention and teacher quality.

Road map. The remainder of the paper is organized as follows. Section 2 describes the Chilean school system and the Carrera Docente reform. Section 3 develops a simple conceptual framework to guide the empirical analysis. Section 4 presents the data and the construction of the analysis samples. Section 5 estimates the overall effect of the reform on teacher retention and examines heterogeneity by teacher value-added. Section 6 tests whether the career stage bonus contributes to retention at the margin. Section 7 concludes.

2 The teaching career

In this section, I describe the main features of the teaching profession in Chile before the introduction of the new salary scheme, and how these features changed under the new system. I focus on aspects of the previous setup that help to understand how the scheme may affect teachers' decisions to stay in the profession.

Pay scheme. Teachers used to earn substantially lower wages than individuals with comparable educational attainment employed in other sectors. Although this pattern is observed in the majority of OECD countries (OECD, 2023), it was particularly pronounced in the Chilean context. For example, Tincani (2021) argues that, conditional on education, earnings outside the teaching profession were approximately 60% higher.[8]

The mixed nature of Chile's school system, with coexisting public and private providers, introduces some variation in teacher compensation across school types. Schools in Chile can be broadly categorized into three types based on their funding and governance: public, private subsidized (or voucher), and private institutions. In 2014, the teaching workforce was distributed approximately equally between public and subsidized schools (45% each), with the remaining 10% employed in private schools.

[8]Therefore, it is not surprising that the teaching career was relatively unattractive to university students. In fact, they tended to be negatively selected in terms of academic performance, as measured by university admission standardized test scores: students with a larger set of options regarding university programs were significantly less likely to choose a career in teaching (Gallegos et al., 2022; De Falco et al., 2024).

Teachers working in public schools were paid according to a rigid pay scale established by the national Teacher Statute, which exclusively rewarded seniority. In contrast, private subsidized and private schools had greater discretion in setting salaries (Behrman et al., 2016). Despite this flexibility, average salaries across public and voucher schools were comparable, with the former exhibiting steeper earnings growth due to the legally mandated seniority-based increments.

Monitoring and bonuses. In the early 2000s, Chile implemented a national teacher evaluation system. A central component of this system was a mandatory portfolio assessment, introduced in 2003. The portfolio required teachers to submit a written unit plan for approximately eight lessons, an example of a student assessment, a video-recorded classroom lesson, and written reflections on their teaching.[9] In 2004, the government introduced the AVDI (*Asignación Variable por Desempeño Individual*), a voluntary assessment available only to teachers who performed sufficiently well in the portfolio.[10] The AVDI consisted of a standardized written test evaluating subject-matter expertise knowledge, using a combination of multiple-choice and short-answer formats tailored to the teacher's area of specialization.[11] It was considered generally not worth taking by a large share of teachers, given the monetary return associated. Both the portfolio evaluation and the AVDI were available only to teachers in public schools; teachers in voucher-funded or private schools were not subject to these evaluations.

2.1 Introduction of the *Carrera Docente*

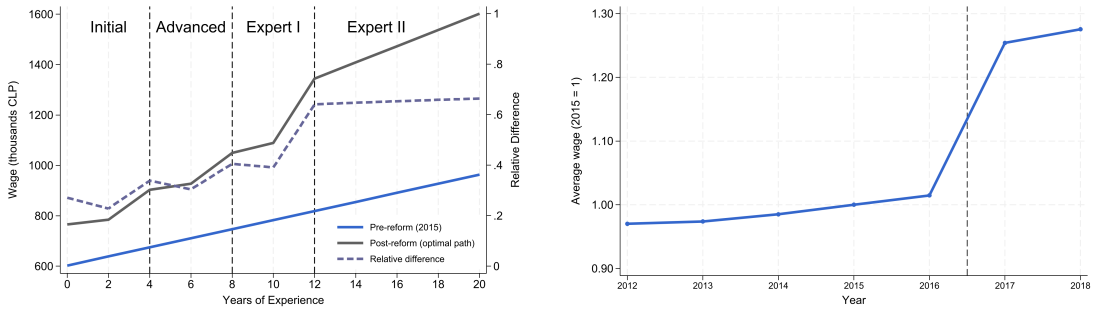
In 2016, Chile enacted Law 20.903, creating the *Carrera Docente*, or *Teacher Professional Development System* (TPDS), a nationwide reform to structure public and voucher school teachers' careers. The system comprised a five-tier career ladder (Initial, Early, Advanced, Expert I, Expert II), preceded by a provisional entry stage (Access).

[9] Teachers were placed into one of four performance categories: *Unsatisfactory*, *Basic*, *Competent* or *Outstanding*. The portfolio was evaluated along with a peer interview, a report from the school principal or supervisor, and a self-evaluation, with the results synthesized into a final rating. However, the largest weight (60%) to define the performance was given to the portfolio. While the evaluation framework formally included the possibility of sanctions—such as dismissal following two consecutive "Unsatisfactory" ratings—evidence suggests these were rarely applied, and the vast majority of teachers received ratings of "Competent" or higher. As such, the system was often perceived more as a monitoring mechanism than as a tool for high-stakes accountability.

[10] They would have to be rated "Competent" or "Outstanding."

[11] Successful candidates received a bonus equivalent to 15% to 25% of their base salary, paid quarterly over two to four years depending on their score. The salary bonus would therefore correspond to a monthly increase of roughly 5–7%. The take-up of the AVDI was around 50%.

Figure 1: Salary before and after the reform



(a) Salary by experience

(b) Average wage over time

Note: Panel (a) shows wages before and after the reform as a function of experience, assuming the teacher advances to the next career stage as soon as eligible. The dashed line shows the relative wage difference between the two schemes. Panel (b) shows the average wage across public school teachers in the analysis sample, normalized to 2015 (see Appendix D for details).

Moreover, the reform introduced an unconditional baseline salary increase of approximately 20% on average, reaching about 30% for teachers at the start of their career.[12] It also linked salary increases to career progression, where each stage corresponds to a specific set of bonuses composed of a fixed component and an experience-based supplement. In contrast to the previous system, salary increases are no longer determined solely by years of experience. Instead, they are contingent upon both teaching experience and performance in two evaluation instruments: the Portfolio and the ECEP (*Evaluación de Conocimientos Específicos y Pedagógicos*). The Portfolio remains structurally similar to its predecessor in the pre-2016 evaluation system and captures pedagogical practice. The ECEP, a standardized test assessing subject-matter knowledge, replaces the earlier AVDI. Unlike its predecessor, the ECEP is mandatory.

Advancement to a higher stage requires meeting both a minimum number of years of teaching and minimum threshold scores on the Portfolio and ECEP. Teachers with sufficient experience could, in principle, access the uppermost stage (Expert II), whereas early-career teachers, even those with high evaluations, may be restricted to intermediate stages such as Advanced. A minimum of four years of experience is required to access the Early and Advanced stages, eight years for Expert I, and twelve

[12]The increase comes from the *Bonificación de Reconocimiento Profesional* (BRP), a bonus granted to all teachers holding a *título profesional* from an accredited higher education institution, which the reform raised from $61,538 to $224,861 per month for a 30-hour contract. In the analysis sample, 96.8% of teachers graduated from a university, instituto profesional, or escuela normal, and are therefore eligible for it. Because the BRP is a flat amount, it is proportionally larger for teachers with lower base pay, which is why the percentage increase is largest at the start of the career.

years for Expert II.[13] Evaluations and advancement take place on a four-year cycle, so that only a fraction of the workforce is assessed in any given year. A teacher who fails to meet the thresholds for the next stage must remain at her current level until the following evaluation round. While promotion up to Advanced is mandatory, further progression beyond this point is voluntary.

Figure 1 plots the old and new salary schemes as a function of years of experience, assuming the teacher advances to each subsequent stage at the earliest opportunity. A teacher who does so could eventually earn a salary nearly 70% higher than under the previous system.[14]

Phase-in. The initial implementation of the reform was restricted to public schools. Voucher schools were incorporated in a staggered way: the first cohort (765 subsidized schools, or 13% of all subsidized schools) was selected in 2017 and entered the system in July 2019.[15] A second cohort of 786 schools was selected in 2018 and entered in July 2020. Evaluations followed a four-year rotation cycle, and this was maintained after the reform. The first cohort to be assessed under the new system in 2016 therefore consisted of teachers who had last been evaluated in 2012.

Figure 2 summarizes the timing of the reform. The first cohort took the Portfolio and ECEP in September–October 2016. Results were published and career stages assigned in July 2017, at which point salary increases and stage-specific bonuses began.[16] A second cohort was evaluated in September–October 2017, with results published and stages assigned in July 2018.

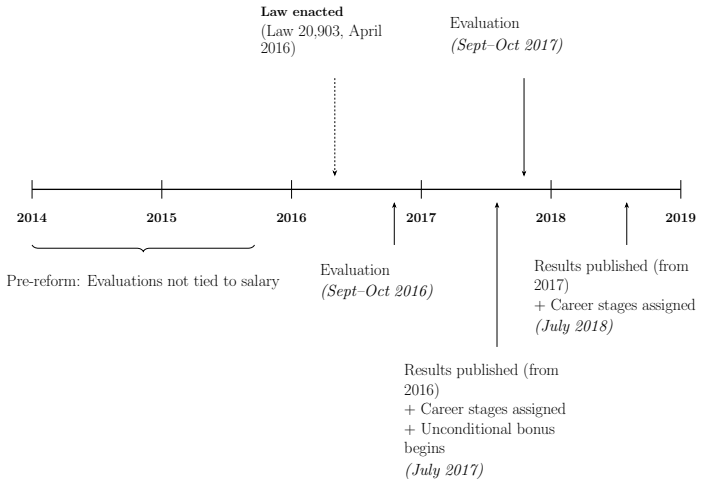
[13] Table B7 illustrates how different combinations of scores and experience determine access to specific career stages.

[14] See Table B1 for a more detailed overview of the bonuses.

[15] The incorporation of voucher schools proceeded in two stages. During the first, voluntary stage (2017–2025), school operators could apply to enter the system, subject to an annual quota set by the Ministry of Education. When applications exceeded available slots, priority was given to schools serving a higher share of disadvantaged students. All remaining voucher schools must enter in a mandatory stage beginning in 2026.

[16] Because not all teachers could be tested immediately, the reform incorporated a transitional mechanism (*asimilación de tramo*) based on pre-existing evaluation results. Teachers' most recent Portfolio scores were used when available, and the AVDI was allowed to substitute for the ECEP. Approximately 45% of teachers who used this mechanism were assigned a career stage in 2017, distributed as 16% Early, 52% Advanced, 23% Expert I, and 8% Expert II.

Figure 2: Timeline of the reform



3 Conceptual framework

This section develops a simple model of teacher exit decisions. The purpose of the model is to guide the empirical analysis, aiming to describe the main interaction behind the exit decision and the structure of the new career. Moreover, it helps to understand how the reform may differentially affect retention across the ability distribution.

Consider a population of public school teachers indexed by teaching effectiveness, or ability, $a \in \mathbb{R}$. Each teacher makes a binary decision: *stay* in public teaching or *exit* to an outside option. Under the pre-reform regime, the teaching wage is $w^{pre} = w_0$, independent of ability or performance.[17] The reform introduces two components:

$$w^{post}(s) = w_0 + \underbrace{\Delta}_{\text{base salary increase}} + \underbrace{b(s)}_{\text{career stage bonus}},$$

where $\Delta > 0$ is a uniform salary increase for all public teachers, s is an evaluation score, and $b(s)$ is a bonus that depends on career stage placement. Career stages are determined by a threshold rule: teachers with $s \geq \tau$ are assigned to a higher stage and receive a bonus $B > 0$. The evaluation score is a noisy measure of teaching effectiveness, $s = a + \nu$, where ν is measurement error independent of a. A teacher with ability a qualifies for the bonus with probability $1 - F_\nu(\tau - a)$, which is increasing in a: more effective teachers are more likely to reach a higher career stage, but the mapping is

[17] In practice, pre-reform wages depend on seniority. I abstract from this for clarity; what matters is that pre-reform wages do not depend on evaluated performance.

imperfect.

Each teacher has an outside option with value $\omega(a)$, with evidence suggesting that more able teachers face better outside options (Bacolod, 2007; Nagler et al., 2020; Chingos and West, 2012). Teachers have strictly concave utility over income, $u(\cdot)$ with $u' > 0$ and $u'' < 0$, and teaching provides a non-pecuniary benefit $\mu = \bar{\mu}(a) + \varepsilon$, where $\bar{\mu}(a)$ captures systematic differences in intrinsic motivation across the ability distribution, for which there is also some evidence (Boyd et al., 2008; Goldhaber et al., 2011), and ε is an idiosyncratic preference shock with logistic distribution $\Lambda(\cdot/\sigma)$. A teacher stays if the total value of teaching exceeds the outside option, $u(w) + \bar{\mu}(a) + \varepsilon \geq u(\omega(a))$, so the exit probability is:

$$h(a, w) = \Lambda\left(\frac{u(\omega(a)) - u(w) - \bar{\mu}(a)}{\sigma}\right). \tag{1}$$

Define the *net outside value* $z(a, w) \equiv \left[u(\omega(a)) - u(w) - \bar{\mu}(a)\right]/\sigma$, so that $h = \Lambda(z)$. Teachers with higher z are more likely to exit. Since the reform raises wages by at least $\Delta > 0$ for all public teachers and utility is strictly increasing, it reduces z and therefore exit. The size of the reduction depends on how close the teacher is to the exit margin and on how much utility the increase buys. Whether the reform retains high- or low-ability teachers more, however, is ambiguous. The model identifies two channels.

Baseline exit rate. The framework predicts that the same wage increase produces a larger reduction in exit probability for teachers who are closer to the exit margin. The treatment effect is therefore concentrated among teachers with higher baseline exit rates.[18] Which group this is depends on the balance between outside options $\omega(a)$ and non-pecuniary benefits $\bar{\mu}(a)$. If outside options improve sufficiently with ability ($u'(\omega(a))\omega'(a)$ large), high-ability teachers have higher baseline exit rates and respond more to the salary increase, consistent with the standard Roy model prediction (Lazear, 2000; Brown and Andrabi, 2023). If instead more effective teachers derive greater intrinsic satisfaction from teaching ($\bar{\mu}'(a) > 0$ large enough), low-ability teachers are closer to the exit margin and the reform disproportionately retains them.

Differential exposure to the career stage bonus. Because evaluation scores are positively correlated with ability, high-ability teachers are more likely to receive the career stage bonus, and their average utility over income is higher. All else equal, this channel

[18] This holds as long as the baseline exit rate is below one half, which is the case for every group in the sample. See Appendix C for the derivation.

would favor the retention of high-ability teachers. Two features of the model work against it. If utility is concave, a given salary increase is worth less to teachers who have already received the base salary increase. And if high-ability teachers have low baseline exit rates, the same increase produces a smaller reduction in their exit probability (see Appendix C for a formal derivation).

The sign of the ability gradient is therefore an empirical question, governed by the relative strength of these channels.

4 Data and samples

4.1 Data

I use administrative data from multiple sources linked at the individual level.

Teachers. I rely on administrative records covering the universe of teachers working in primary and secondary education from 2004 to 2024, provided by the Ministry of Education (MINEDUC). These data contain detailed information on teachers' employment status, such as the type of contract and total hours worked. I also observe information on the school where teachers are employed, including whether it is public, subsidized, or private. From 2017 onward, the data additionally report the teacher's career stage within the TPDS. Moreover, I observe teacher evaluation outcomes: the AVDI and the Portfolio assessments are available from 2003 onward, while results from the ECEP are available from 2016. The data do not contain information on salaries. However, because public school teachers are paid according to a statutory salary schedule, I can impute wages from the observed variables in the data (experience, contracted hours, teaching level, career stage, and degree type). Details are provided in Appendix D.

Students. I match teachers to the universe of their students and observe student performance on standardized tests. These data come from SIMCE (*Sistema de Medición de la Calidad de la Educación*), a national assessment administered by the Agencia de Calidad de la Educación. SIMCE tests are administered annually to students in selected grades, and cover mathematics and reading in every year. For each student, I also observe a rich set of sociodemographic characteristics collected through a survey administered together with the test, including parental education and household income.

The analysis sample imposes two restrictions. First, I exclude private school teachers and subsidized school teachers employed in schools that entered the Carrera Docente before 2020, to avoid capturing behavioral responses of teachers in voucher schools that already anticipate the introduction of the new career. Second, I restrict the sample to teachers under the age of 50 to avoid confounding exit from the profession with retirement.[19]

4.2 Measurement

Measuring exit rates. A central challenge in studying teacher attrition is defining when a teacher has genuinely left the profession. One way to do it is to define an indicator at time t equal to 1 if the teacher is not observed in the data for year $t+1$. However, this definition confounds exit with temporary absences such as parental leave or sabbaticals. In the sample, 7.5% of teachers under 50 are not observed in the following year.[20] Yet among these one-year absentees, over half (50.3%) reappear in the data within five years. This suggests that the one-year measure may substantially overstate genuine attrition. Throughout the paper, I therefore use a five-year exit measure: a teacher is classified as having exited at time t if she is not observed in any of years $t+1$ through $t+5$. This more conservative definition captures persistent exit while allowing for temporary absences, and reduces the average exit rate to 3.7%.

Appendix Figure A2 plots exit rates by years of experience for public and subsidized school teachers. Exit rates in subsidized schools are generally higher than in public schools across the experience distribution.[21] Among teachers with fewer than five years of experience, the five-year exit rate averages approximately 6%, compared to 2–3% for those with 5 to 15 years. Comparing panels (a) and (b) reveals that the one-year exit measure exhibits a similar shape but at higher levels throughout.

[19]The first restriction removes approximately 40,000 private school teachers (13% of the workforce) and approximately 56,000 subsidized school teachers in the 1,509 subsidized schools observed in the data, out of the 1,551 that were incorporated into the career system in July 2019 or July 2020 (18% of the workforce). The age restriction removes an additional 86,000 teachers (27% of the workforce). The estimation window ends in 2018. Exit measured in year t is the decision to leave between t and $t+1$, so 2018 is the last year in which that decision is taken before the pandemic. Extending the window to 2019 would measure departures between 2019 and 2020, which cannot be separated from the effects of Covid-19.

[20]This figure is comparable to the annual attrition rates reported for other countries: approximately 8% of public school teachers in the United States leave the profession each year (National Center for Education Statistics, 2023), 9–10% in England, and 6.5% on average across OECD countries (OECD, 2025).

[21]Teachers in public and subsidized schools are subject to different labor legislation. Public school teachers are governed by the *Estatuto Docente* (Teacher Statute), which provides strong employment protections, including restrictions on dismissal. Teachers in subsidized schools, on the other hand, are employed under the *Código del Trabajo* (Labor Code), the standard private-sector labor regulation, which gives employers greater flexibility in hiring and firing decisions.

Measuring teacher ability: value-added in test scores. To measure teaching effectiveness, or ability, I estimate teacher value-added (TVA) from student performance on the SIMCE standardized tests in mathematics and reading. SIMCE is administered annually, but coverage varies across grades and years.[22] I restrict the estimation sample to teachers observed in grades 6, 8, and 10, where I can link each teacher to the students in her classroom and observe both current and lagged test scores.

The TVA estimation follows a standard value-added model, estimated separately for mathematics and reading:

$$\text{TestScore}_{sct} = \beta\,\text{TestScore}_{sc,t-1} + \gamma\,\text{Female}_s + C'_{ct}\delta + \theta_{k(c)} + \lambda_{gt} + \alpha_{it} + \epsilon_{sct}, \qquad (2)$$

where TestScore_{sct} is the standardized SIMCE score of student s in classroom c at time t; $\text{TestScore}_{sc,t-1}$ is the student's lagged score in the same subject; C_{ct} is a set of variables related to classroom characteristics (class size, share female, lagged class average score, and parental education and income quintiles); $\theta_{k(c)}$ are school type fixed effects (public, private, technical and rural); and λ_{gt} are grade-by-year fixed effects. The teacher-by-year fixed effects α_{it}, where i is the teacher in year t, are the TVA estimates; they identify each teacher's contribution to student test score growth in year t, net of student, classroom, and school characteristics.

4.3 Three analysis samples

The empirical analysis proceeds in three steps, each relying on a different sample. First, I evaluate the overall effect of the career system on teacher retention using the full sample of public and subsidized school teachers described above, exploiting the fact that the reform was initially restricted to the public sector. Appendix Table B3 presents descriptive statistics for this sample, split by school type and reform period. Women account for roughly three quarters of the teaching workforce, and the average teacher is about 35 years old with 8 years of experience. Public school teachers are more likely to work in rural areas (21% vs. 4%) and serve a more disadvantaged student body.

Second, to assess whether the reform differentially retains high- or low-ability teachers, I restrict attention to the subset of teachers for whom I can estimate pre-reform value-added. Because TVA requires linking teachers to SIMCE-tested classrooms, this

[22] Appendix Table B2 reports the full SIMCE testing schedule by grade and year. I estimate TVA using the universe of teachers linked to SIMCE-tested classrooms, without imposing the age or sector restrictions of the DiD sample.

measure is available only for a fraction of the workforce. More precisely, I restrict to teachers whose value-added is estimated in 2013 or 2014.[23] This yields a subsample of approximately 19,000 teachers, or about 15 percent of the full analysis sample. Appendix Table B4 in the appendix compares the two groups: they are broadly similar in demographic and school characteristics, though teachers with TVA exhibit a somewhat lower five-year exit rate (1.8 versus 3.9 percent).

Third, to examine whether the additional career stage bonus contributes to retention at the margin, I focus on the subset of public school teachers who were evaluated under the new system in 2016 or 2017. As described in Section 2, teachers are called for evaluation on a regular four-year cycle, so the evaluated population in any given year is determined by the rotation schedule rather than by self-selection. I describe this sample in detail and compare it to the full population of public school teachers in Section 6.

5 Salary increases and teacher retention

In this section, I first examine the overall effect of introducing the base salary increases, and bonuses corresponding to career stages, on teacher retention. Then, I perform a heterogeneity analysis based on pre-determined TVA to characterize which teachers are more likely to respond.

5.1 Empirical strategy

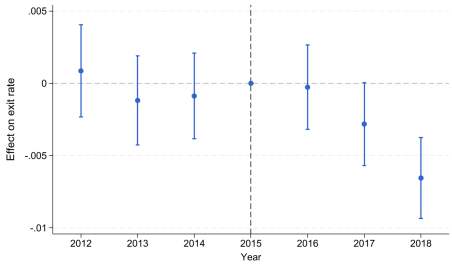
I estimate the following event study specification:

$$\text{Exit}_{ist} = \alpha_s + \sum_{t \neq 2015} \beta_t \left(\text{Public}_{it} \times \mathbf{1}[\text{Year} = t]\right) + \gamma_t + \delta(\text{Exp}_{it}) + \varepsilon_{ist} \tag{3}$$

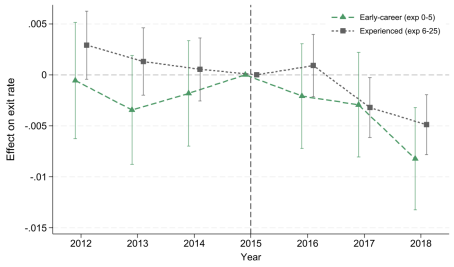
where Exit_{ist} is an indicator for whether teacher i in school s exits the profession at time t, defined as not being observed in any of years $t+1$ through $t+5$. Public_{it} is an indicator for teachers employed in a public school in year t. Because the reform applies at the school level, treatment status follows the school a teacher works in, and can therefore change if she moves across sectors. α_s, γ_t, and $\delta(\text{Exp}_{it})$ are school fixed effects, year fixed effects, and experience fixed effects, respectively. The coefficients β_t represent the differential change in exit rates for public school teachers relative to subsidized school teachers, normalized to zero in 2015, the last pre-reform year. Standard errors

[23] Araya-Córdova et al. (2025) show that the reform changed teaching practices and effort. Therefore, comparing teachers based on their productivity post-reform may be problematic.

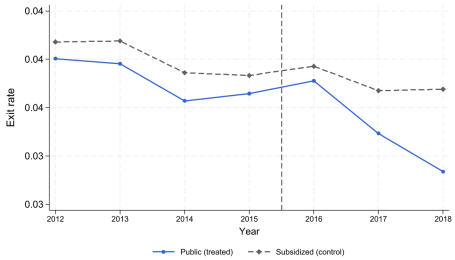
Figure 3: Effect of the career system on teacher exit



(a) All teachers



(b) By experience



(c) Raw exit rates

Note: Event study coefficients β_t from equation (3) with 95% confidence intervals. Baseline year: 2015. Panel (a) shows the full sample. Panel (b) splits by early-career (0–5 years of experience) and experienced (6+ years) teachers. Panel (c) shows raw mean exit rates by school type. Teachers under 50 in public and subsidized schools, 2012–2018. Standard errors clustered at the school $\times$ year level.

are clustered at the school $\times$ year level. Causal interpretation of β_t requires that, absent the reform, exit rates in public and subsidized schools would have followed parallel trends, the standard assumption in this setting.

5.2 Average effect on exit

Figure 3 presents the main results. Panel (a) reports the event study coefficients from equation (3) for the full sample. The effect is close to zero in 2016, the year in which the first cohort of teachers was evaluated but had not yet received any salary adjustment. It becomes detectable in 2017, when career stages were assigned and the associated salary increases began, and continues to grow in 2018. By 2018, public school teachers are 0.7 percentage points less likely to exit than subsidized school teachers relative to the pre-reform period.

Given the large baseline differences in exit rates across the experience distribution, I estimate the event study separately for early-career (0–5 years of experience) and

experienced (6+ years) teachers, reported in panel (b). Early-career teachers exit at a rate of approximately 6% before the reform, while the rate among experienced teachers is 2–3%. Despite these different baselines, I detect a significant decrease in exit rates for both groups, amounting to roughly 15% relative to the baseline for early-career teachers and 20% for experienced teachers. The reform, therefore, appears to have been effective in increasing retention broadly, including among experienced teachers, whose baseline exit rates were already low.

Using imputed wages from the statutory salary schedule (Appendix D), I can quantify the implied elasticity. The average imputed wage for public teachers increased by approximately 25 percent between 2015 and 2017 (Figure 1b). Combined with an 18 percent reduction in exit (0.7 percentage points on a 3.7 percent baseline), this implies an elasticity of exit with respect to wages of approximately -0.8. While this implicitly treats the salary increase as permanent, if teachers anticipate that voucher schools will enter the system the elasticity is approximately -1.8.[24] There is however evidence that teachers heavily discount future compensation (Biasi, 2019; Fitzpatrick, 2015). I therefore take -0.8 as the reference value, with -1.8 as an upper bound. Because the reform restructured compensation for the entire public teaching workforce, this elasticity is directly informative for the design of large-scale salary policies. Existing estimates based on broad wage variation are scarce and come from indirect sources, such as cross-municipal differences in Norway (-1.3, Falch 2011) or seniority-driven pay growth in the United States (-1.4, Hendricks 2014). The larger elasticities of -3 to -8 found for targeted retention bonuses (Clotfelter et al., 2008; Springer et al., 2016; Tartova, 2025; Cowan and Goldhaber, 2018) are less relevant for predicting the effect of an across-the-board raise, since compliers in those settings are in hard-to-staff schools.

In specification (3) treatment is defined at the school by year level. The estimates may therefore conflate two margins. The first is a behavioral response. The second is selection into public schools. Teachers who are marginally attracted to a public school may differ in unobserved ways that correlate with their propensity to leave the profession. While this is not a problem for identification as long as the object of interest is the average exit rate of the school, the second margin is however likely to play a minor role. In fact, after the reform only about 8 percent of public school teachers each year are

[24] Voucher schools enter the system between 2019 and 2026 on a schedule set by the law. Discounting the wage differential over the realized entry cohorts at 3 percent, and comparing it with earnings over a remaining career of 20 years, the 25 percent increase corresponds to about 10.8 percent of the present value of earnings. Dividing the 18 percent reduction in exit by this figure gives -1.8.

new hires, and movements across sectors are limited.[25] Moreover, when performing the same analysis on incumbent teachers,[26] results are similar to the ones shown in Figure 3, with the notable exception that the effect on exit rates is estimated negative also in 2016, the year in which teachers had been evaluated but not yet paid (Appendix Figure A3a).

5.3 Threats to identification

The pre-reform coefficients in both panels are small and statistically indistinguishable from zero, pointing in the direction that exit rates in public and subsidized schools were evolving similarly before the reform. This is further confirmed by the raw exit rates in panel (c): the exit rate among subsidized school teachers remains roughly constant throughout the sample period, while it drops for public school teachers after 2016.

During the same period, Chile implemented some reforms in the education system.[27] At the school level, the most significant concurrent policy change was the Ley de Inclusión (Law 20,845), which replaced school-level admissions in subsidized schools with a centralized lottery system (the Sistema de Admisión Escolar, SAE). The stated goal of the SAE was to reduce socioeconomic segregation in the subsidized sector by eliminating selective admissions. To the extent that the reform changed the composition of students in subsidized schools, it could in principle have affected teacher retention, since student composition is a well-documented determinant of teacher turnover (Hanushek et al., 2004; Ronfeldt et al., 2013). However, the available evidence indicates that the SAE had limited effects on student sorting (Kutscher et al., 2023; Elacqua et al., 2025). Moreover, in Appendix E I directly test whether the SAE rollout affected teacher exit rates by exploiting its staggered regional implementation in a difference-in-differences design, and I find no evidence in favor of this hypothesis.

Wage spillovers. Voucher schools may have responded to the reform by raising their own wages, which would attenuate the estimated effect (Tincani, 2021). The wage increase was large and difficult to match for schools that did not receive the state transfers funding the reform. The flat trend in exit rates among subsidized school teachers visible

[25] In 2016, for example, only 1,893 teachers moved into a public school from a subsidized one, compared with 5,320 hired into public schools with no prior experience. The share of subsidized school teachers moving to the public sector is 3.6 percent in 2015, 4.1 percent in 2016 and 4.0 percent in 2018.

[26] I assign $\text{Public}_i = 1$ to teachers working in a public school the first year in which they are observed in the sample, fixing the baseline year at 2012, so that treatment is pre-determined and unaffected by compositional changes induced by the reform.

[27] See, for example, De Falco and Reichlin (2025) and Bucarey (2018) on the expansion of financial aid, and De Falco (2025) and Gallegos et al. (2022) on targeted scholarships for prospective teachers.

in panel (c) of Figure 3 indirectly suggests no such response. In Appendix F, I provide a more direct test using household survey data (CASEN) and find that composition-adjusted real wages for private sector teachers were essentially unchanged between 2013 and 2017, while those for public teachers increased by approximately 18%, consistent with the increase estimated using imputed wages.

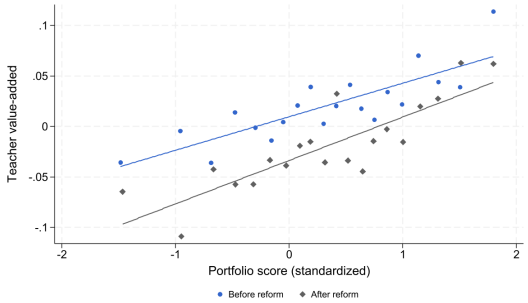
Additional robustness. Finally, I test the robustness of these results in two relevant ways. First, Appendix Figure A4a shows that the event study pattern is virtually identical when using one- through four-year exit measures instead of the five-year measure used in the main specification. Second, Appendix Figure A4b shows that the estimates are stable when progressively adding the teacher and school characteristics described in Appendix Table B3.

Who gets retained?

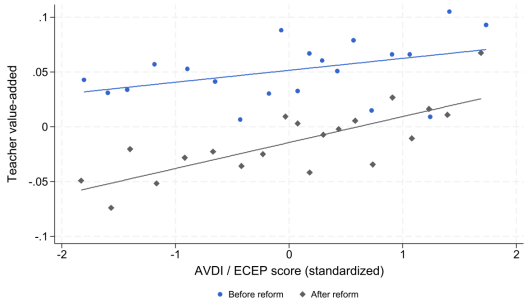
Before turning to the heterogeneity analysis, I discuss two mechanisms of the conceptual framework using the TVA estimates from Section 4.2.

TVA and the evaluations. A central assumption of the conceptual framework is that evaluation scores are positively correlated with teaching effectiveness. This relationship is what generates differential exposure to the career stage bonus, since higher-ability teachers are more likely to score above the evaluation thresholds and reach higher career stages, raising their expected post-reform wage. Whether this assumption holds empirically is therefore important for interpreting the heterogeneity analysis that follows. Appendix Table B5 reports OLS coefficients from regressions of TVA on standardized evaluation scores, matching each evaluation observation to the closest TVA measured within three years (see Figure 4, panels a and b, for a non-parametric representation). A one standard deviation increase in the Portfolio score is associated with an increase of 0.03 in TVA, and the relationship is stable across the pre- and post-reform periods. The AVDI and ECEP show similar coefficients. To put these magnitudes in perspective, the standard deviation of teacher value-added is typically estimated at 0.10 to 0.15 (Bacher-Hicks, 2022). A coefficient of 0.03 therefore represents roughly one-quarter of the cross-teacher dispersion in effectiveness, a magnitude comparable to those found in studies that directly correlate classroom observation protocols or teacher knowledge assessments with value-added (Rockoff et al., 2011; Bacher-Hicks et al., 2019).

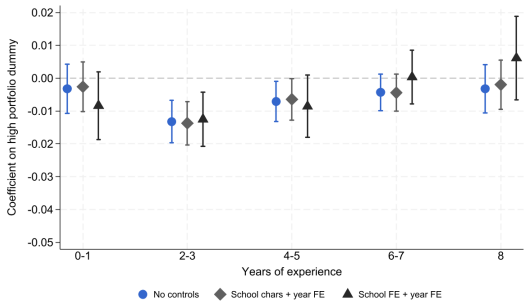
Figure 4: Ability, evaluation scores and pre-reform exit



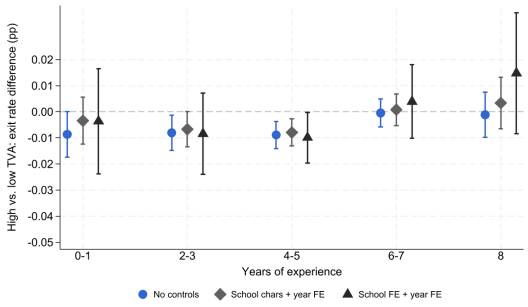
(a) TVA and Portfolio score



(b) TVA and AVDI or ECEP score



(c) High Portfolio score and exit



(d) High TVA and exit

Note: Panels (a) and (b) show binned scatter plots of teacher value-added (TVA) against standardized evaluation scores, estimated separately before the reform (2012–2015) and after the reform (2016–2018). Each evaluation observation is matched to the closest TVA measured within three years. Panel (a) uses Portfolio scores, panel (b) uses AVDI scores before the reform and ECEP scores after. Panels (c) and (d) show the coefficient on a high-ability indicator (above the within-experience-bin median) from regressions of five-year exit within two-year experience bins in the pre-reform period (2012–2015), using Portfolio scores in panel (c) and combined TVA in panel (d). 95% confidence intervals based on robust standard errors.

Ability and exit rates. The conceptual framework highlights that the sign of the treatment effect gradient also depends on which teachers are closer to the exit margin. To assess this empirically, I examine how pre-reform exit rates vary with ability. Figure 4, panels c and d, reports, for both Portfolio scores and TVA, the coefficient on a high-ability indicator from regressions of exit rates within two-year experience bins, under three different specifications (i) no controls, (ii) school characteristics with year fixed effects, and (iii) school fixed effects with year fixed effects. Across all specifications, the coefficient is negative at mid-experience levels, indicating that teachers with higher evaluation scores or higher TVA are less likely to exit.

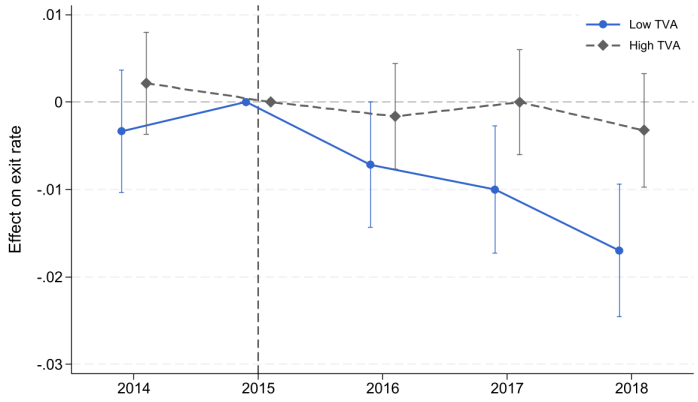
Together, these results indicate that both measures of ability are negatively associ-

ated with baseline exit, consistent with the conceptual framework's scenario where the non-pecuniary channel dominates the outside option channel. This is also consistent with evidence from the United States, where less effective teachers are generally found to be more likely to leave (Hanushek et al., 2004; Boyd et al., 2008; Goldhaber et al., 2011).

5.4 Effect of the reform by teacher ability

As mentioned before, this analysis focuses on the subset of teachers for whom I can estimate pre-reform teacher value-added (TVA). Each teacher is assigned their most recent pre-treatment TVA and is tracked forward from that measurement year. I classify teachers as high or low TVA based on the within-sector and within-subject median in the pre-reform period. The distribution of TVA is similar in the two sectors (Appendix Figure A5).

Figure 5: Effect on exit by pre-reform teacher value-added



Note: Event study coefficients β_t from equation (3) estimated separately for teachers above (high TVA) and below (low TVA) the within-sector median of pre-reform TVA. Baseline year: 2015. Standard errors clustered at the school $\times$ year level. 95% confidence intervals shown.

TVA can be measured either in 2013 or 2014, but 72% of the sample has its most recent TVA in 2014. To avoid an unbalanced first year in which only the smaller 2013 cohort contributes, I start the panel in 2014. Teachers whose TVA is measured in 2013 are tracked from 2014 onward, alongside those measured in 2014. As a robustness check, I also restrict to the 2014 cohort only.

I estimate equation (3) separately for teachers above and below the within-sector

TVA median. The retention effect of the reform appears concentrated among low-TVA teachers. Figure 5 presents the event study estimates separately for the two groups, showing that the post-reform decline in exit rates is visibly present for teachers below the median. Table 1 summarizes this pattern by reporting the pooled post-reform DiD coefficient for each group. Across all specifications, the point estimate for low-TVA teachers is larger in absolute value than for high-TVA teachers. When pooling both cohorts and defining post as 2016–2018 (column 1), the effect is -1.10 percentage points ($p < 0.01$) for low-TVA teachers versus -0.16 percentage points for high-TVA teachers, not distinguishable from zero. The gap is larger when restricting the post period to 2017–2018 (columns 3 and 4).

Table 1: Effect on teacher exit by ability level

	Post = 2016–2018		Post = 2017–2018	
	Both cohorts (1)	2014 only (2)	Both cohorts (3)	2014 only (4)
Low TVA	-0.0110***	-0.0117***	-0.0133***	-0.0133***
	(0.0031)	(0.0035)	(0.0033)	(0.0037)
High TVA	-0.0016	-0.0006	-0.0015	-0.0004
	(0.0026)	(0.0029)	(0.0027)	(0.0031)
Difference (High − Low)	0.0094**	0.0111**	0.0118***	0.0128***
	(0.0040)	(0.0045)	(0.0043)	(0.0048)
N (Low TVA)	37,295	28,167	37,295	28,167
N (High TVA)	37,885	28,764	37,885	28,764

Note: Each cell reports the coefficient on Post × Public from a difference-in-differences regression of the 5-year exit rate on year fixed effects, a public school indicator, their interaction, and experience controls, with school fixed effects and standard errors clustered at the school × year level. The baseline year is 2015. Teachers are classified as high or low TVA using subject-specific within-sector medians. Value added is estimated separately for mathematics and reading, and 5.0% of teachers have it in both subjects. For them the mathematics estimate is used. *Both cohorts* pools teachers with TVA measured in 2013 and 2014; *2014 only* restricts to the 2014 cohort. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Robustness. Threats to the DiD design, such as concurrent reforms and wage spillovers, were addressed in Section 5 and apply to this subsample as well. The pre-reform coefficient in Figure 5 is close to zero for both groups, indicating that exit rates in public and subsidized schools were evolving similarly not only on average but also within each TVA group. Moreover, the results are robust to restricting to the 2014 TVA cohort only (Table 1, columns 2 and 4) and to adding the teacher and school characteristics from

Appendix Table B3 (Appendix Figure A6).[28] A concern specific to this analysis is that measurement error in TVA could affect the classification of teachers into high and low ability, so that the differential effect depends on the particular value-added specification used. In Appendix Figure A7, I show the estimates by re-estimating TVA with different sets of controls, always keeping the lagged test score. Notably, coefficients are stable, and so is the difference in effects between low and high TVA.

As documented above, low-TVA teachers have higher baseline exit rates. The larger retention response among them is what the framework predicts when the teachers closest to the exit margin are the low-ability ones. This finding contrasts with some recent evidence from France Tartova (2025), where wage bonuses targeted at teachers in disadvantaged schools differentially retain better teachers. I moreover test whether the larger retention response among low-TVA teachers holds within low- and high-performing schools. More precisely, I split each TVA group at the within-sector median of the pre-reform school SIMCE score and estimate equation (3) separately in each of the four groups. Appendix Figure A8 shows that the effect I estimate on average for low- and high-TVA teachers is very similar to the one I estimate in these subgroups.

Yet this pattern may appear puzzling in light of the reform's design. Since TVA and evaluation scores are positively correlated (Appendix Table B5), high-ability teachers can expect larger career stage bonuses, which should, if anything, strengthen their incentive to stay. That the retention effect nonetheless concentrates among low-ability teachers raises the question of whether the career stage bonus contributes to retention at all. The next section aims to shed light on this question directly.

6 The marginal effect of the performance bonus

In this section I examine whether the career stage bonus, the skill-based component of the reform, contributes to retention at the margin. Among teachers who have already received the baseline salary increase, I estimate the marginal impact of an additional bonus using a regression discontinuity design that exploits the cutoff rules governing career stage assignment.

[28] Appendix Table B6 compares low- and high-TVA teachers within each sector. High-TVA teachers are slightly more experienced, have longer tenure at their school, and teach in schools with higher SIMCE scores.

6.1 Regression discontinuity design

As described in Section 2, the assignment of career stages depends on two evaluation scores, Portfolio and ECEP, conditional on experience (Table B7). This creates a two-dimensional threshold structure with multiple margins at which a teacher can advance to a higher stage (e.g., Early to Advanced, Advanced to Expert I, and Expert I to Expert II). To make the analysis as transparent as possible, I proceed in two steps.

First, I focus on a single margin: the transition from Early to Advanced. I therefore restrict the sample to teachers with at least four years of experience, the minimum required for eligibility to the Advanced stage. This choice is motivated by two considerations. On the one hand, the Early-to-Advanced boundary concentrates the largest mass of evaluated teachers, providing greater statistical power to detect an effect (see Appendix Figure A9 for the joint distribution of scores). On the other hand, it allows me to study the broadest population of teachers, since higher transitions require stricter minimum experience thresholds, while still capturing a non-negligible jump in compensation (Table B1). Second, I reduce the two-dimensional assignment problem to a single running variable.[29] I construct a single running variable by recentering each teacher's Portfolio score at the relevant Early-to-Advanced threshold conditional on their ECEP category.[30] Teachers with a positive recentered score fall above the cutoff and are assigned to the Advanced stage; those with a negative value remain in the Early stage.

I estimate the following regression discontinuity model:

$$Exit_{i,t+1} = \beta_0 + \beta_1 \cdot \mathbb{1}[\widetilde{P}_{it} \geq 0] + f(\widetilde{P}_{it}) + X_i'\gamma + u_i, \tag{4}$$

where $Exit_{i,t+1}$ is the exit outcome for teacher i in year $t+1$. Teachers are evaluated in year t but learn their career stage assignment and begin receiving the associated salary adjustment only in year $t+1$ (Figure 2), so this is the first year in which the exit decision

[29] For a discussion of multi-score regression discontinuity designs, see Papay et al. (2011) and Reardon and Robinson (2012).

[30] The Portfolio cutoff for the Early-to-Advanced transition is 2.50 for teachers with ECEP in $[1.88, 2.75)$, 2.25 for ECEP in $[2.75, 3.38)$, and 2.00 for ECEP $\geq$ 3.38. Teachers with an ECEP score below 1.88 are excluded from the RD sample, since no Early-to-Advanced Portfolio cutoff is defined for them. I use the Portfolio rather than the ECEP as the running variable because the ECEP score distribution exhibits bunching at certain scale values, including at career stage thresholds (Appendix Figure A1, panel b). The Portfolio, by contrast, is scored by trained evaluators on multiple rubric indicators and the final score is a weighted average, yielding a much smoother distribution. The bunching in the ECEP violates the smoothness assumption required for using it as a running variable, but does not affect the RDD along the Portfolio dimension, as confirmed by the density continuity test presented below.

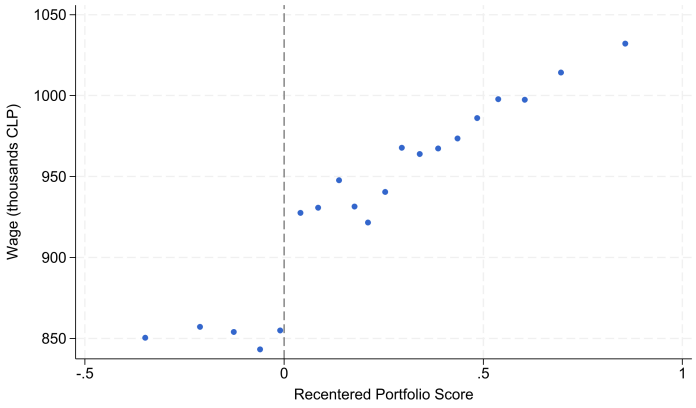
can be affected. A teacher is classified as having exited if she is not observed in any of years $t+2$ through $t+k+1$, where k is the exit horizon. $\widetilde{P}_{it}$ is the recentered Portfolio score from the year-t evaluation, $f(\cdot)$ is a local polynomial estimated separately on each side of the cutoff, and X_i is a vector of pre-determined covariates. Bandwidths are chosen optimally following Calonico et al. (2019). The parameter of interest is β_1, which captures the causal effect of crossing the Early-to-Advanced threshold on teacher exit at the cutoff.

RD sample. Roughly 20,000 public school teachers with at least four years of experience were evaluated in 2016 or 2017. Appendix Table B8 compares them to the full population of public school teachers. Evaluated teachers are somewhat older (mean age 37.5 vs. 35.0) and more experienced (10.5 vs. 7.5 years), as expected given the minimum experience requirement. Once conditioning on at least four years of experience, however, the two populations are very similar across all observable characteristics, consistent with the four-year rotation schedule determining who is evaluated rather than self-selection.

Identification. The identifying assumption underlying the RDD is that potential outcomes vary smoothly through the cutoff, so that any discontinuity in the outcome can be attributed to the change in career stage assignment. I test this assumption through the standard set of validity checks. First, I test for manipulation of the running variable using the density continuity test of McCrary (2008). Appendix Figure A10 shows no evidence of a discontinuity in the density of the recentered Portfolio score at the cutoff, suggesting that teachers are unable to precisely manipulate their score to cross the threshold. Second, I verify that pre-determined covariates are balanced around the cutoff. As shown in Appendix Table B9, only one of the thirteen RD estimates for teacher or school characteristics is statistically different from zero at conventional levels, the share of mothers with college education in the school, which is in line with what would be expected by chance.

Figure 6 displays the first stage. Wages jump by approximately 95,000 CLP per month, about 11 percent, at the cutoff of the recentered Portfolio score. Teachers scoring just below the threshold can still reach the Advanced stage at the following evaluation round. The jump is nonetheless considerable. Using the estimated probability of passing at the next evaluation for those just below the threshold, around 50 percent, it is equivalent in present value terms to a 10 percent higher salary for approximately 6.5

Figure 6: Wage at the Early-to-Advanced threshold



Note: Binscatter of monthly wages (thousands of CLP) against the recentered Portfolio score. Wages are computed from the statutory salary schedule (Appendix D).

years.[31]

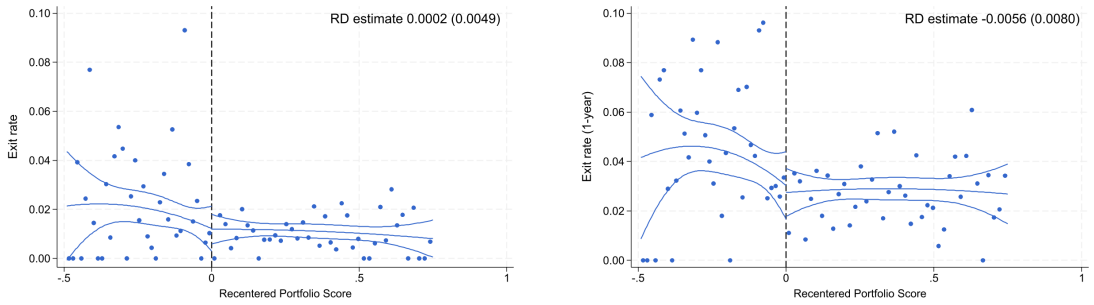
6.2 The effect of receiving an additional bonus

Figure 7 plots exit rates as a function of the recentered Portfolio score, using both the five-year and one-year exit measures. The figures show no visible discontinuity at the cutoff for either outcome. Exit rates are flat through the threshold, with the quadratic fits and confidence intervals overlapping on both sides. The estimated effect on exit is close to zero and never statistically distinguishable from zero for both the five-year and the one-year measures. The estimates are virtually unchanged when adding covariates (Appendix Table B10) and across bandwidths (Appendix Figure A13).

As with any regression discontinuity design, this estimate is local to teachers near the threshold. Several features of the setting, however, suggest that the complier population is not unusually selected. The relevant Portfolio cutoffs fall squarely in the middle of the score distribution (Appendix Figure A1, panel a). Moreover, because evaluations take place on a four-year rotation, the RD sample captures a broad cross-section of the teaching workforce rather than a self-selected group. Consistent with this, the observable characteristics of evaluated teachers are similar to those of the overall

[31]The discontinuous jump in wages reflects the assignment to the Advanced stage at the cutoff (Appendix Figure A11). Not all teachers above the cutoff are assigned to the Advanced stage, and not all teachers below remain in the Early stage, because of the transitional mechanism (*asimilación de tramo*) that provisionally assigned some teachers to a higher stage based on pre-existing evaluation results. Appendix Figure A12 shows the probability of assignment to each career stage as a function of both Portfolio and ECEP scores.

Figure 7: Exit rates around the Early-to-Advanced threshold



(a) Five-year exit

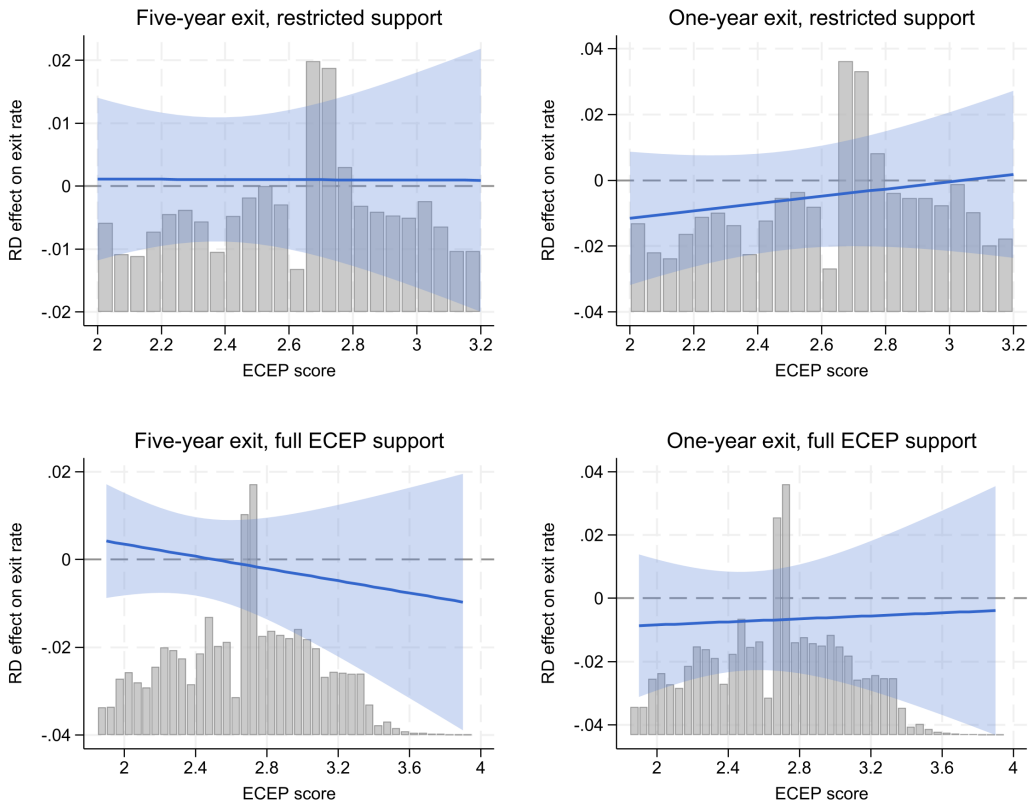
(b) One-year exit

Note: Binned means of exit rates as a function of the recentered Portfolio score at the Early-to-Advanced threshold. Panel (a) uses five-year exit; panel (b) uses one-year exit. Quadratic fits with 95% confidence intervals estimated separately on each side of the cutoff.

population of public school teachers. Finally, given the two-dimensional assignment rule of the bonus, the discontinuity here is a frontier rather than a single point. The effect can therefore be estimated at different points along the frontier, interacting the indicator variable in (4) with the ECEP score (Papay et al., 2011). I estimate this interaction both on the full support of the ECEP score and on a slightly restricted support, which excludes the long but sparsely populated upper tail, where observations may have high leverage on the slope. Results are shown in Figure 8. In both cases, the estimated effect stays close to zero over the whole support of the ECEP score. The estimates at each point are imprecise, but the absence of a statistically significant gradient suggests that the null does not hide an effect concentrated among a group of teachers.

Taken together, the salary increase substantially reduced exit, while the career stage bonus paid on top of it has no detectable effect at the margin. Because the *Carrera Docente* bundles both components, this setting allows the marginal contribution of a performance-based bonus to be isolated among teachers who have already received a large unconditional raise. Previous studies could only infer nonlinear retention effects indirectly, for example, from the fact that turnover elasticities decline with experience as seniority raises wages (Hendricks, 2014), or from the observation that retention bonuses of very different sizes produce similar effects across contexts (Springer et al., 2016; Clotfelter et al., 2008).

Figure 8: Effect of crossing the Portfolio threshold along the ECEP support



Note: Effect of crossing the Early-to-Advanced Portfolio threshold on exit, estimated as a function of the ECEP score within the MSE-optimal bandwidth, with the discontinuity interacted linearly with the ECEP score. Solid lines show the estimated effect and shaded areas the 95% confidence intervals. Grey bars show the distribution of the ECEP score among teachers within the bandwidth. Left panels use the five-year exit measure and right panels the one-year measure. Top panels restrict the sample to teachers with an ECEP score up to 3.2, bottom panels use the full ECEP support.

7 Conclusion

This paper studies the combined effect of unconditional salary increases and performance-based bonuses on teacher retention and workforce composition, using Chile's *Carrera Docente* as a natural experiment. Using a DiD design, I show that the reform as a whole reduced teacher exit by approximately 18%, with an implied elasticity of -0.8. This effect on retention, however, is concentrated among low-value-added teachers. Their exit rate decreases significantly, while the effect for high-value-added teachers is indistinguishable from zero. I rationalize this heterogeneity through the lens of a Roy model, where teachers trade off pecuniary and non-pecuniary characteristics of the profession

when deciding to leave. The pattern is consistent with less effective teachers being closer to the exit margin, and therefore more responsive to the salary increase, because high-ability teachers derive greater non-pecuniary satisfaction from the profession. Finally, I use a regression discontinuity design to isolate the performance-based bonus, and find that it has no detectable effect on retention.

While these findings may suggest a trade-off between retention and teacher quality, it is worth noting that this is a short-term evaluation of this type of compensation. Unconditional salary increases are generally found not to increase the productivity of teachers (De Ree et al., 2018), and for this reform in particular Araya-Córdova et al. (2025) show that the skill bonus component has only a small effect on effort, affecting a small share of teachers. Given the results shown in this paper, one may therefore conclude that this trade-off exists. However, one should also consider longer-term outcomes. A higher salary combined with skill-based bonuses may change the decision of prospective teachers to enter the profession, and if the candidates it attracts are better teachers than the incumbents, the compositional effect documented here could be offset over time.

References

Araya-Córdova, P., Contreras, D., Sepúlveda, P., and Rodríguez, J. (2025). Getting teachers back to study: Input-based teacher incentives and student achievement. Working paper, Universidad de los Andes.

Bacher-Hicks, A., Chin, M. J., Kane, T. J., and Staiger, D. O. (2019). An experimental evaluation of three teacher quality measures: Value-added, classroom observations, and student surveys. *Economics of Education Review*, 73:101919.

Bacher-Hicks, K. (2022). Estimation and interpretation of teacher value added in research applications. *Handbook of the Economics of Education 6, 93-134*.

Bacolod, M. P. (2007). Do alternative opportunities matter? The role of female labor markets in the decline of teacher quality. *Review of Economics and Statistics*, 89(4):737–751.

Bates, M. D. and Johnston, A. C. (2025). Do pensions enhance teacher effort and selective retention? Working Paper 33986, National Bureau of Economic Research.

Behrman, J. R., Tincani, M. M., Todd, P. E., and Wolpin, K. I. (2016). Teacher quality in public and private schools under a voucher system: The case of chile. *Journal of Labor Economics*, 34(2):319–362.

Berlinski, S. and Ramos, A. (2020). Teacher mobility and merit pay: Evidence from a voluntary public award program. *Journal of Public Economics*, 186:104186.

Biasi, B. (2019). Higher salaries or higher pensions? inferring preferences from teachers' retirement behavior. Working Paper, Yale School of Management.

Biasi, B. (2021). The labor market for teachers under different pay schemes. *American Economic Journal: Economic Policy*, 13(3):63–102.

Bobba, M., Ederer, T., León-Ciliotta, G., Neilson, C., and Nieddu, M. G. (2021). Teacher compensation and structural inequality: Evidence from centralized teacher school choice in Perú. Working Paper 29068, NBER.

Boyd, D., Grossman, P., Lankford, H., Loeb, S., and Wyckoff, J. (2008). Who leaves? teacher attrition and student achievement. Working Paper 14022, National Bureau of Economic Research.

Brown, C. and Andrabi, T. (2023). Pay, preferences, and the teacher labor market: How financial incentives affect teacher sorting. *Journal of Political Economy*, 131(6):1578–1622.

Bucarey, A. (2018). Central admissions in school choice: Evidence from chile's college admission system. Working paper, MIT.

Busso, M., Montaño, S., Muñoz-Morales, J. S., and Pope, N. G. (2024). The unintended consequences of merit-based teacher selection: Evidence from a large-scale reform in Colombia. *Journal of Public Economics*, 239:105088.

Calonico, S., Cattaneo, M. D., and Farrell, M. H. (2019). Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs. *The Econometrics Journal*, 23(2):192–210.

Cattaneo, M. D., Jansson, M., and Ma, X. (2020). Simple local polynomial density estimators. *Journal of the American Statistical Association*, 115(531):1449–1455.

Chetty, R., Friedman, J. N., and Rockoff, J. E. (2014). Measuring the impacts of teachers i: Evaluating bias in teacher value-added estimates. *American economic review*, 104(9):2593–2632.

Chingos, M. M. and West, M. R. (2012). Do more effective teachers earn more outside the classroom? *Education Finance and Policy*, 7(1):8–43.

Clotfelter, C., Glennie, E., Ladd, H., and Vigdor, J. (2008). Would higher salaries keep teachers in high-poverty schools? evidence from a policy intervention in north carolina. *Journal of Public Economics*, 92(5-6):1352–1370.

Cowan, J. and Goldhaber, D. (2018). Do bonuses affect teacher staffing and student achievement in high-poverty schools? *Economics of Education Review*, 65:138–152.

Dal Bó, E., Finan, F., and Rossi, M. A. (2013). Strengthening State Capabilities: The Role of Financial Incentives in the Call to Public Service. *The Quarterly Journal of Economics*, 128(3):1169–1218.

De Falco, A. (2025). Teacher scholarships and teacher supply. Working paper, University of Bologna.

De Falco, A., Hattemer, B., and Sierra Vásquez, S. (2024). Recruiting better teachers? evidence from a higher education reform in chile. *Available at SSRN: https://ssrn.com/abstract=4874361 or http://dx.doi.org/10.2139/ssrn.4874361*.

De Falco, A. and Reichlin, Y. (2025). *Grants vs. Loans: the Role of Financial Aid in College Major Choice*. SSRN working paper available at https://ssrn.com/abstract=4581235 or http://dx.doi.org/10.2139/ssrn.4581235.

De Ree, J., Muralidharan, K., Pradhan, M., and Rogers, H. (2018). Double for nothing? experimental evidence on an unconditional teacher salary increase in indonesia. *The Quarterly Journal of Economics*, 133(2):993–1039.

Dee, T. S. and Keys, B. J. (2004). Does merit pay reward good teachers? Evidence from a randomized experiment. *Journal of Policy Analysis and Management*, 23(3):471–488.

Dee, T. S. and Wyckoff, J. (2015). Incentives, selection, and teacher performance: Evidence from IMPACT. *Journal of Policy Analysis and Management*, 34(2):267–297.

Deneault, C. (2025). Local labor markets and selection into the teaching profession. *Working Paper*.

Dinerstein, M. and Opper, I. M. (2022). Screening with multitasking: theory and empirical evidence from teacher tenure reform. Technical report, National Bureau of Economic Research.

Duflo, E., Hanna, R., and Ryan, S. P. (2012). Incentives work: Getting teachers to come to school. *American Economic Review*, 102(4):1241–78.

Elacqua, G., Kutscher, M., et al. (2025). Navigating centralized admissions: The role of parental preferences in school segregation in chile. *Economics of Education Review*.

Falch, T. (2011). Teacher mobility responses to wage changes: Evidence from a quasi-natural experiment. *American Economic Review*, 101(3):460–465.

Finan, F., Olken, B. A., and Pande, R. (2017). The personnel economics of the developing state. *Handbook of economic field experiments*, 2:467–514.

Fitzpatrick, M. D. (2015). How much are public school teachers willing to pay for their retirement benefits? *American Economic Journal: Economic Policy*, 7(4):165–188.

Fraenkel, R. C. (2022). Local labor markets and job match quality: Teachers. *Labour Economics*, 78.

Gallegos, S., Neilson, C., Calle, F., and Karnani, M. (2022). *Screening and Recruiting Talent At Teacher Colleges Using Pre-College Academic Achievement*. Princeton University, Industrial Relations Section.

Glewwe, P., Ilias, N., and Kremer, M. (2010). Teacher incentives. *American Economic Journal: Applied Economics*, 2(3):205–227.

Goldhaber, D., Gross, B., and Player, D. (2011). Teacher career paths, teacher quality, and persistence in the classroom: Are public schools keeping their best? *Journal of Policy Analysis and Management*, 30(1):57–87.

Hanushek, E. A., Kain, J. F., and Rivkin, S. G. (2004). Why public schools lose teachers. *Journal of Human Resources*, 39(2):326–354.

Hendricks, M. D. (2014). Does it pay to pay teachers more? evidence from texas. *Economics of Education Review*, 40:79–90.

Holmstrom, B. and Milgrom, P. (1991). Multitask principal–agent analyses: Incentive contracts, asset ownership, and job design. *The Journal of Law, Economics, and Organization*, 7(Special Issue):24–52.

Ingersoll, R. M. (2001). Teacher turnover and teacher shortages: An organizational analysis. *American Educational Research Journal*, 38(3):499–534.

Jackson, C. K. (2018). What do test scores miss? the importance of teacher effects on non–test score outcomes. *Journal of Political Economy*, 126(5):2072–2107.

Johnston, A. C. (2025). Preferences, selection, and the structure of teacher pay. *American Economic Journal: Applied Economics*, 17(3):310–346.

Kutscher, M., Nath, S., and Urzúa, S. (2023). Centralized admission systems and school segregation: Evidence from a national reform. *Journal of Public Economics*, 221:104864.

Lavy, V. (2009). Performance pay and teachers' effort, productivity, and grading ethics. *American Economic Review*, 99(5):1979–2011.

Lazear, E. P. (2000). Performance pay and productivity. *American Economic Review*, 90(5):1346–1361.

Leaver, C., Ozier, O., Serneels, P., and Zeitlin, A. (2021). Recruitment, effort, and retention effects of performance contracts for civil servants: Experimental evidence from rwandan primary schools. *American economic review*, 111(7):2213–2246.

McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics*, 142(2):698–714.

Muralidharan, K. and Sundararaman, V. (2011). Teacher performance pay: Experimental evidence from india. *Journal of political Economy*, 119(1):39–77.

Nagler, M., Piopiunik, M., and West, M. R. (2020). Weak markets, strong teachers: Recession at career start and teacher effectiveness. *Journal of Labor Economics*, 38(2):453–500.

National Center for Education Statistics (2023). Teacher attrition and mobility: Results from the 2021–22 teacher follow-up survey. first look. Technical Report NCES 2024-039, U.S. Department of Education.

OECD (2023). What do oecd data on teachers' salaries tell us? *Education Indicators in Focus*, (83).

OECD (2025). Education at a glance 2025: OECD indicators. Technical report, OECD Publishing, Paris.

Papay, J. P., Willett, J. B., and Murnane, R. J. (2011). Extending the regression-discontinuity approach to multiple assignment variables. *Journal of Econometrics*, 161(2):203–207.

Reardon, S. F. and Robinson, J. P. (2012). Regression discontinuity designs with multiple rating-score variables. *Journal of Research on Educational Effectiveness*, 5(1):83–104.

Rivkin, S. G., Hanushek, E. A., and Kain, J. F. (2005). Teachers, schools, and academic achievement. *Econometrica*, 73(2):417–458.

Rockoff, J. E., Jacob, B. A., Kane, T. J., and Staiger, D. O. (2011). Can you recognize an effective teacher when you recruit one? *Education finance and Policy*, 6(1):43–74.

Ronfeldt, M., Loeb, S., and Wyckoff, J. (2013). How teacher turnover harms student achievement. *American Educational Research Journal*, 50(1):4–36.

Roy, A. D. (1951). Some thoughts on the distribution of earnings. *Oxford Economic Papers*, 3(2):135–146.

Santibáñez, L., Martínez, J. F., Datar, A., McEwan, P. J., Setodji, C. M., and Basurto-Dávila, R. (2007). Breaking ground: Analysis of the assessment system and impact of Mexico's teacher incentive program "Carrera Magisterial". Technical Report TR-458-SEP, RAND Corporation, Santa Monica, CA.

Springer, M. G., Swain, W. A., and Rodriguez, L. A. (2016). Effective teacher retention bonuses: Evidence from Tennessee. *Educational Evaluation and Policy Analysis*, 38(2):199–221.

Tartova, D. (2025). Who leaves and who stays? the impact of the teacher wage gap on teacher quality. Job Market Paper, Paris School of Economics.

Taylor, E. S. and Tyler, J. H. (2012). The effect of evaluation on teacher performance. *American Economic Review*, 102(7):3628–3651.

Tincani, M. M. (2021). Teacher labor markets, school vouchers, and student cognitive achievement: Evidence from chile. *Quantitative Economics*, 12(1):173–216.

Tournier, B. and Chimier, C. (2019). Teacher career reforms in Peru. Technical report, UNESCO International Institute for Educational Planning, Paris.

UNESCO (2023). The teachers we need for the education we want: the global imperative to reverse the teacher shortage. Technical report, United Nations Educational, Scientific and Cultural Organization.

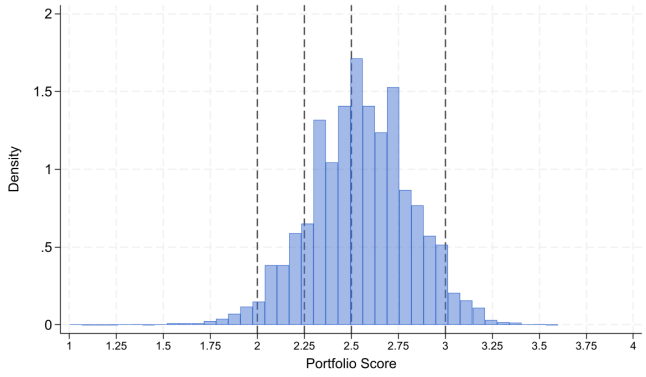
Appendix - Salary Increases and Skills-Based Bonuses: Teacher Retention and Workforce Composition

Appendices

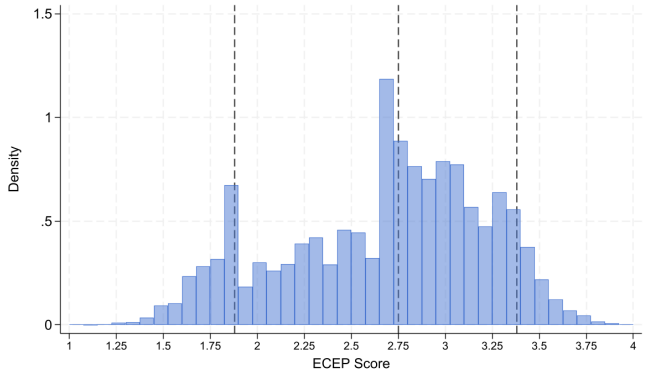
A Additional figures

Figure A1: Distribution of evaluation scores

(a) Portfolio scores

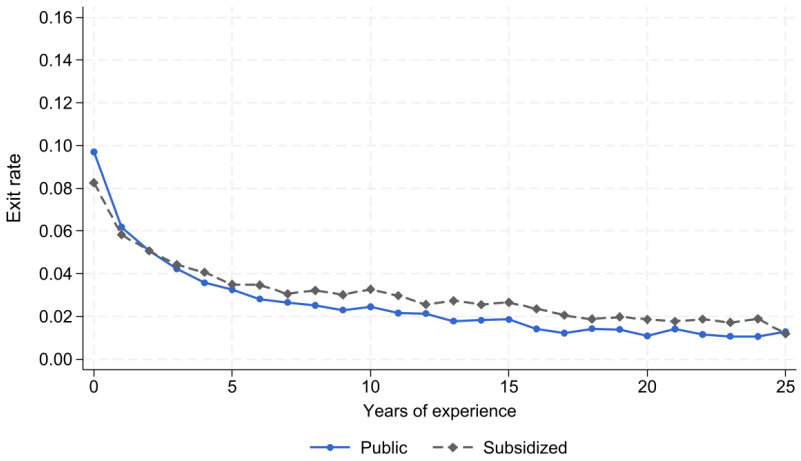


(b) ECEP scores

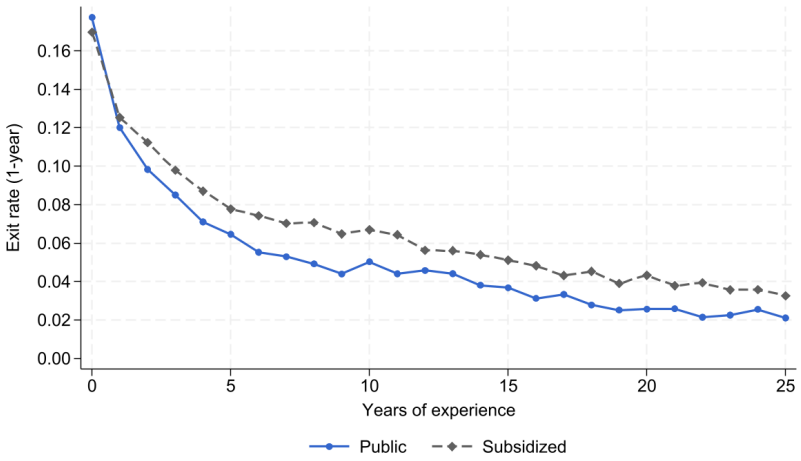


Note: Distribution of Portfolio (top) and ECEP (bottom) scores for public school teachers evaluated in 2016 or 2017 with at least four years of experience. Dashed vertical lines indicate the thresholds determining career stage assignment (Table B7).

Figure A2: Exit rates by experience and school type



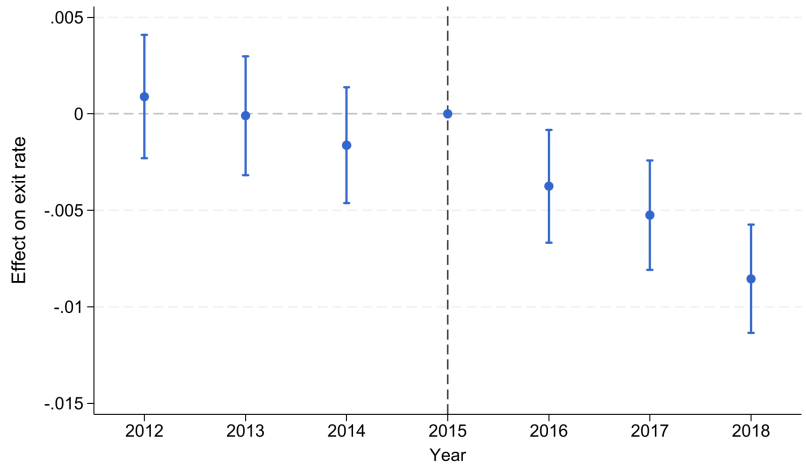
(a) Five-year exit



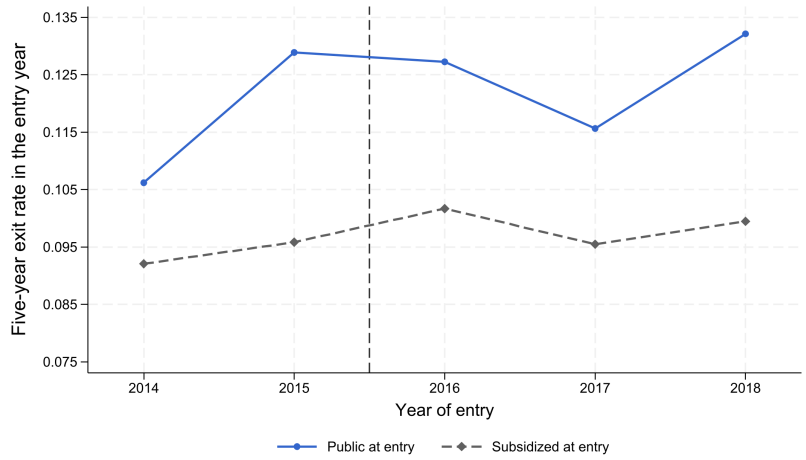
(b) One-year exit

Note: Exit rates by years of experience, separately for public and subsidized school teachers. Panel (a) uses five-year exit; panel (b) uses one-year exit. Sample restricted to teachers under 50, pre-reform period (2012–2015).

Figure A3: Incumbents and new hires



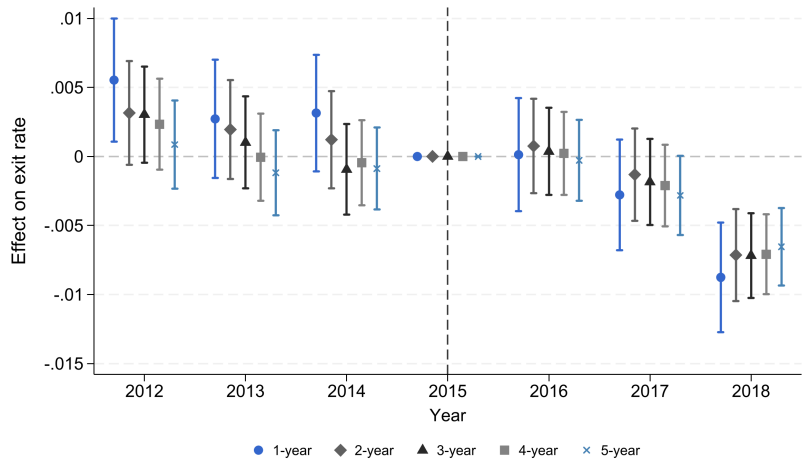
(a) Incumbents with sector fixed at entry



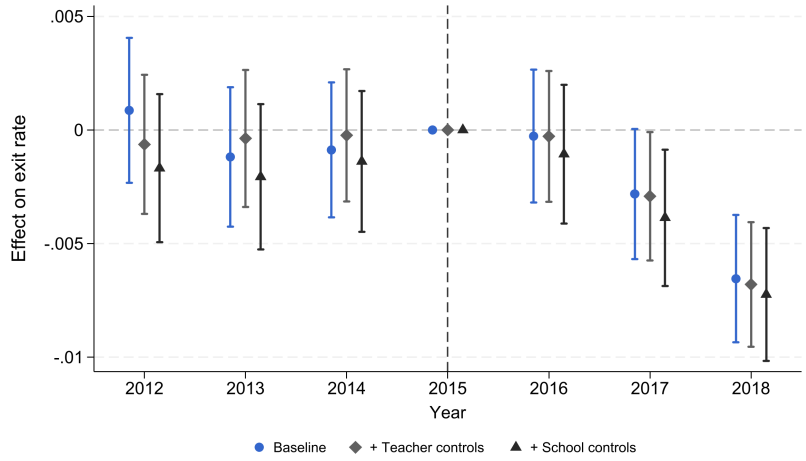
(b) New hires with no prior experience

Note: Panel (a) reports event study coefficients β_t from equation (3) with 95% confidence intervals and 2015 as the omitted year. The sample includes teachers first observed in 2015 or earlier. Each teacher is assigned $\text{Public}_i = 1$ if the first school in which she is observed is a public one, and this assignment is kept for all later years. The sample covers 157,425 teachers and roughly 745,000 teacher-year observations. Standard errors are clustered at the school $\times$ year level. Panel (b) reports the five-year exit rate in the year a teacher is first observed in the sample, among teachers with no prior years of service, by year of entry and by the sector of the school that hires them. Teachers first observed in 2012 are excluded from panel (b), since the panel starts that year.

Figure A4: Robustness of the main event study



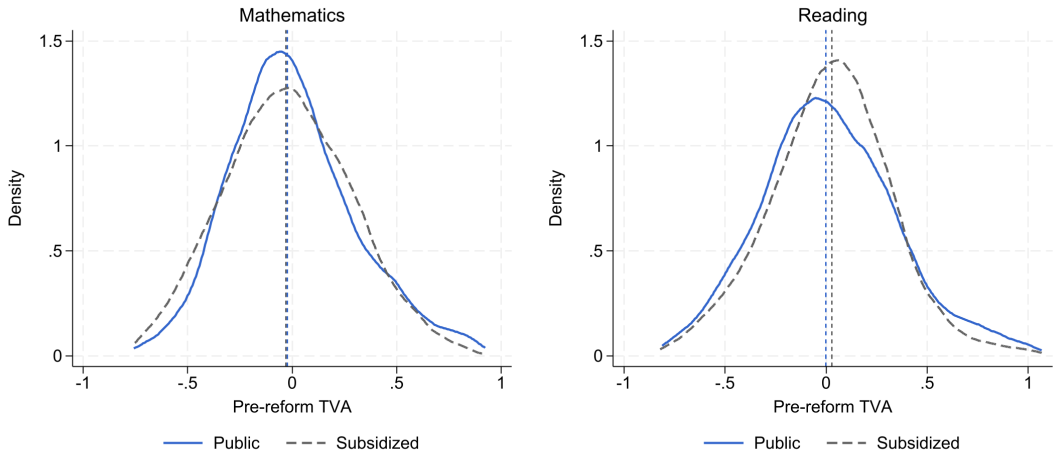
(a) Alternative exit horizons



(b) Progressive controls

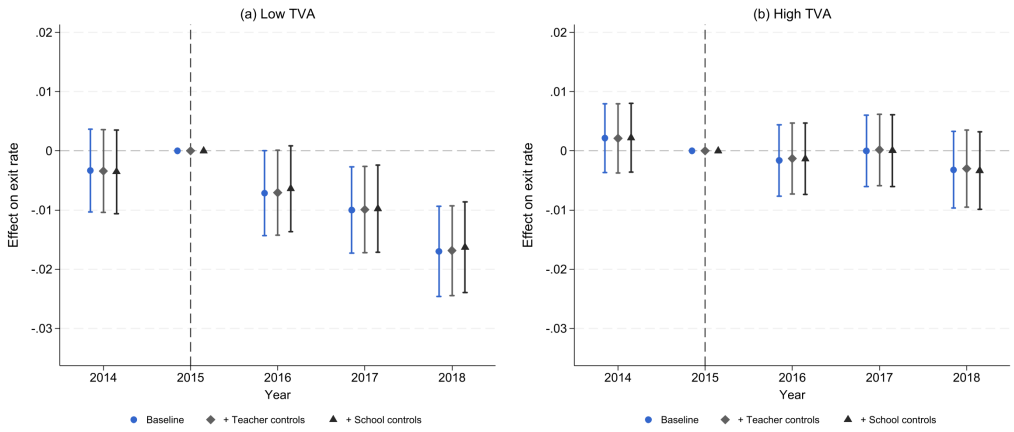
Note: Panel (a) shows event study coefficients β_t from equation (3) using exit measures defined as not being observed in any of years $t+1$ through $t+k$, for $k = 1, \ldots, 5$. The main specification uses $k = 5$. Panel (b) shows estimates with progressively richer controls. "Baseline" includes school fixed effects and experience fixed effects. "+ Teacher controls" adds gender, age, age squared, tenure at current school, university graduation, degree duration, and teaching level indicators (preschool, primary, secondary). "+ School controls" further adds school-level SIMCE scores and the share of mothers with college education. Standard errors clustered at the school $\times$ year level. 95% confidence intervals shown.

Figure A5: Distribution of pre-reform teacher value-added by sector



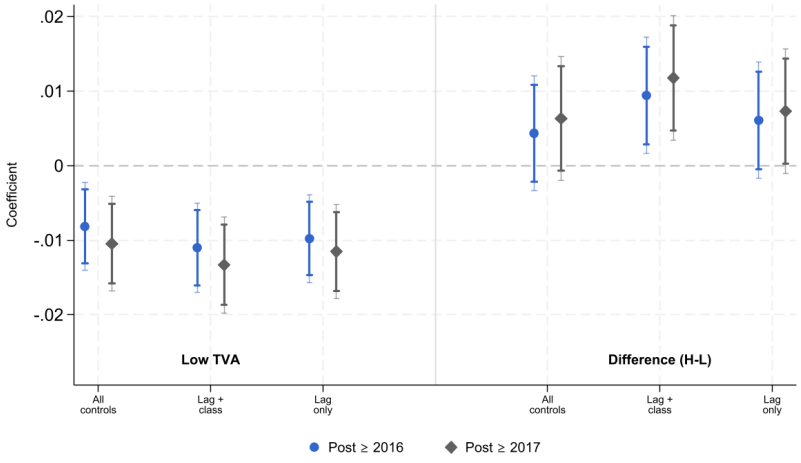
Note: Kernel densities of pre-reform teacher value-added (TVA) in public and subsidized schools, for teachers in the TVA sample observed in 2014. Dashed vertical lines show the within-sector medians used to classify teachers as high or low TVA.

Figure A6: Robustness of TVA heterogeneity to additional controls



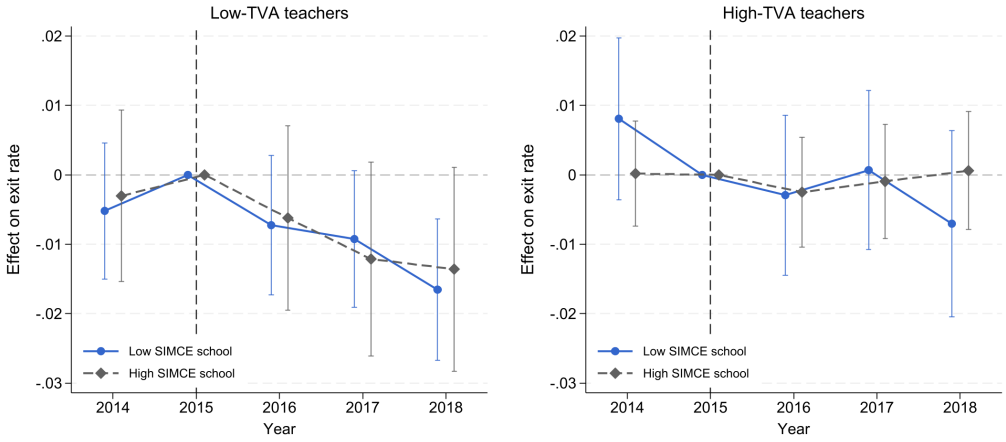
Note: Event study coefficients from equation (3) estimated separately for low-TVA (panel a) and high-TVA (panel b) teachers under three specifications: baseline (school and experience fixed effects), adding teacher controls (gender, age, age squared), and further adding school controls (SIMCE math and reading scores, share of mothers with college education). Teachers are classified using subject-specific within-sector medians. 95% confidence intervals shown. Standard errors clustered at the school $\times$ year level.

Figure A7: Robustness of TVA heterogeneity to alternative estimation strategies



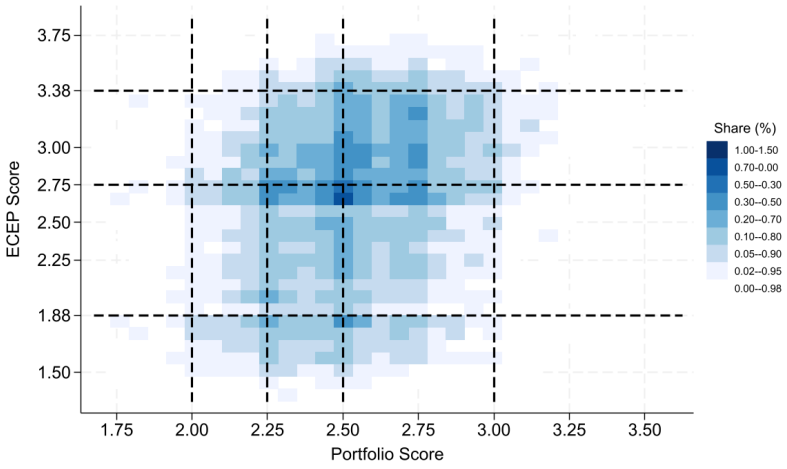
Note: Left panel shows the pooled post-reform DiD coefficient for low-TVA teachers; right panel shows the difference between high- and low-TVA coefficients. Each group of bars corresponds to a different TVA estimation strategy: all controls (lagged score, classroom, and school composition), lagged score and classroom controls (baseline specification), and lagged score only. Teachers are classified using subject-specific within-sector medians. Thick bars show 90% confidence intervals; thin bars show 95% confidence intervals. Post $\geq$ 2016, both cohorts.

Figure A8: Effect by teacher value-added and school performance



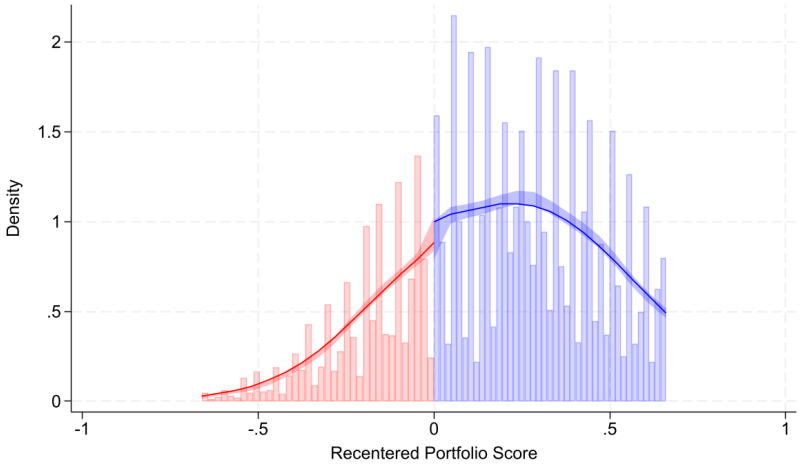
Note: Event study coefficients from equation (3) estimated separately for low-TVA (left) and high-TVA (right) teachers, splitting each group by the school mean SIMCE mathematics score in the year the teacher's value-added is measured, at the within-sector median. Teachers are classified using subject-specific within-sector medians. 95% confidence intervals shown. Standard errors clustered at the school $\times$ year level.

Figure A9: Joint distribution of Portfolio and ECEP scores



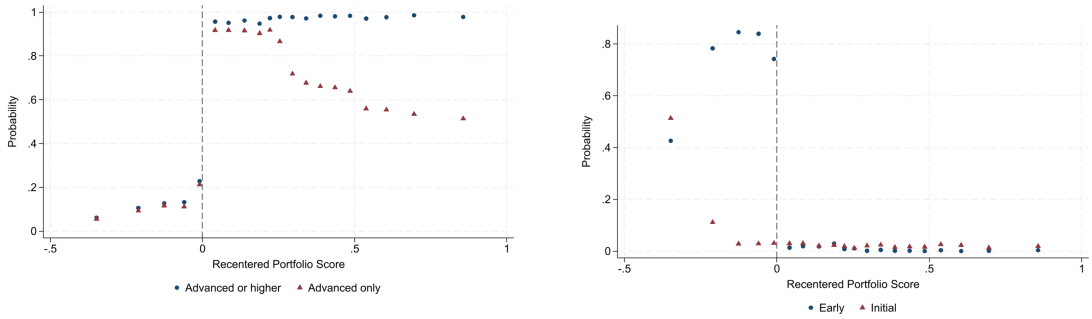
Note: Joint distribution of Portfolio and ECEP scores for public school teachers evaluated in 2016 or 2017 with at least four years of experience. Each cell shows the share of teachers in a 0.05×0.05 score bin. Dashed lines indicate career stage thresholds (Table B7).

Figure A10: Density continuity test



Note: Density continuity test following Cattaneo et al. (2020) for the recentered Portfolio score at the Early-to-Advanced threshold. Histograms show observed frequencies; curves show local polynomial density estimates with pointwise confidence intervals on each side of the cutoff.

Figure A11: Career stage assignment at the threshold

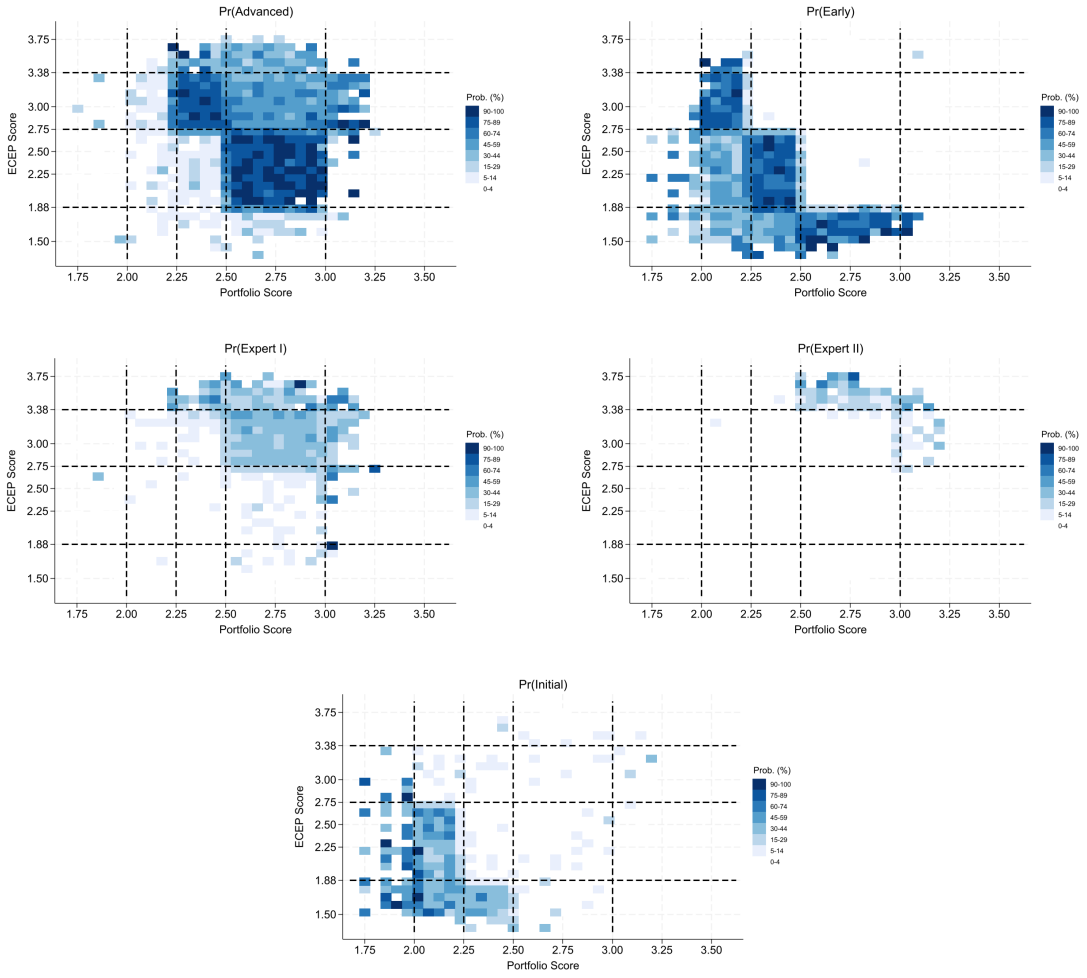


(a) Advanced or higher / Advanced only

(b) Early / Initial

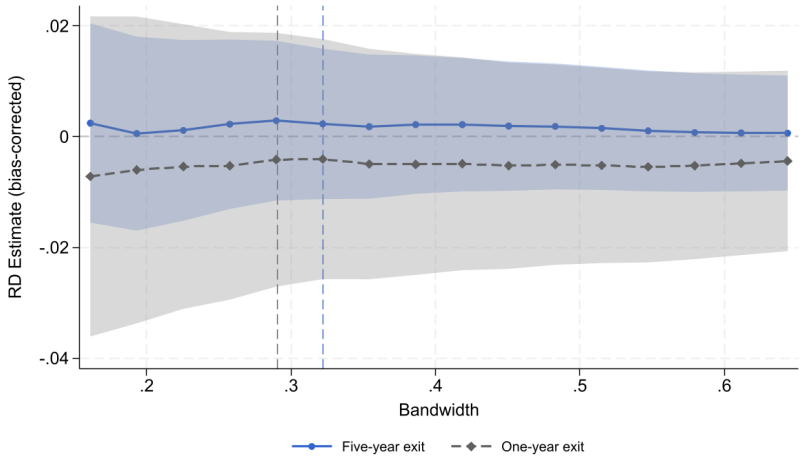
Note: Binscatter of the probability of being assigned to each career stage against the recentered Portfolio score. The running variable is the teacher's Portfolio score minus the ECEP-specific threshold for reaching Advanced. Panel (a) shows the probability of reaching Advanced or higher (circles) and Advanced only (triangles). Panel (b) shows the probability of being assigned to Early (circles) and Initial (triangles). 20 equal-sized bins on each side.

Figure A12: Career stage assignment by Portfolio and ECEP scores



Note: Probability of being assigned to each career stage as a function of Portfolio and ECEP scores. Each cell shows the mean probability in a 0.05×0.05 score bin (bins with fewer than 3 observations excluded). Dashed lines indicate career stage thresholds (Table B7). RD sample: public school teachers evaluated in 2016 or 2017 with at least four years of experience.

Figure A13: Bandwidth sensitivity



Note: Bias-corrected RD estimates of the effect of reaching Advanced on teacher exit, estimated over a grid of bandwidths from 50% to 200% of the MSE-optimal bandwidth. Shaded areas show 95% confidence intervals based on robust standard errors. Vertical dashed lines indicate the MSE-optimal bandwidth for each outcome.

B Additional tables

Table B1: Fixed and variable components of the career stage bonus

Category	**Fixed Bonus (CLP)**	**Variable Bonus (CLP)**
Initial		$13,076 \times \frac{Hrs}{44} \times \frac{b}{15}$
Early		$43,084 \times \frac{Hrs}{44} \times \frac{b}{15}$
Advanced	$90,000 \times \frac{Hrs}{44}$	$86,714 \times \frac{Hrs}{44} \times \frac{b}{15}$
Expert I	$125,000 \times \frac{Hrs}{44}$	$325,084 \times \frac{Hrs}{44} \times \frac{b}{15}$
Expert II	$190,000 \times \frac{Hrs}{44}$	$699,593 \times \frac{Hrs}{44} \times \frac{b}{15}$

Note: Fixed and variable bonuses associated with each career stage, expressed in Chilean pesos (CLP). The amounts are the maxima set by Article 49 of Law 20.903, defined for a 44-hour contract and 15 biennia of experience. *Hrs* refers to the number of contractual working hours, and b to completed biennia of experience, capped at 15.

Table B2: Cohorts taking the SIMCE by year

	Cohort												
Year	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013
2006			4										
2007				4									
2008					4								
2009						4							
2010	10						4						
2011				8				4					
2012		10							4		2*		
2013			10			8		6		4		2*	
2014				10			8		6		4		2*
2015					10			8		6		4	
2016						10					6		4
2017							10			8			
2018								10					6
2019												8	

Note: Cohorts are identified by the year in which students start 1st grade of elementary school. The table shows in which grades different cohorts took the SIMCE. Grades 4, 6, and 8 correspond to elementary school grades, and 10th grade corresponds to high school (2do medio). Second graders are only tested on their Spanish knowledge. I use grades 6, 8, and 10 for the TVA estimation, since the value-added model requires a lagged test score. Fourth grade is excluded because no prior SIMCE score is available.

Table B3: Summary statistics

	Public				Subsidized			
	Pre-reform		Post-reform		Pre-reform		Post-reform	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Panel A: Teacher characteristics								
Female	0.74	0.44	0.74	0.44	0.77	0.42	0.77	0.42
Age	35.5	7.5	35.1	6.9	34.6	7.1	34.6	6.7
Experience (years)	7.8	6.9	7.6	6.3	8.0	7.1	8.1	7.1
Tenure at current school	4.5	5.3	4.2	4.7	4.9	5.5	4.9	5.5
University graduate	0.85	0.36	0.88	0.33	0.87	0.33	0.90	0.29
Degree duration (semesters)	8.6	1.5	8.7	1.4	8.9	1.4	9.0	1.3
Contracted hours (weekly)	35.1	9.1	37.4	8.6	35.2	9.1	36.0	8.8
Primary education	0.61	0.49	0.60	0.49	0.47	0.50	0.47	0.50
Panel B: School characteristics								
Far North	0.06	0.24	0.06	0.24	0.06	0.25	0.06	0.24
North	0.08	0.27	0.08	0.27	0.07	0.26	0.07	0.26
Center	0.50	0.50	0.50	0.50	0.63	0.48	0.62	0.49
South	0.26	0.44	0.27	0.44	0.18	0.38	0.19	0.39
Far South	0.10	0.29	0.10	0.29	0.06	0.24	0.06	0.24
Rural	0.21	0.41	0.21	0.41	0.04	0.20	0.04	0.20
SIMCE math score	238.8	25.2	237.3	25.0	267.6	27.2	266.7	26.9
SIMCE reading score	239.0	20.7	239.0	21.1	259.5	20.5	258.4	19.8
Share mothers with college	0.04	0.05	0.05	0.07	0.15	0.14	0.17	0.14
Teacher-year observations	218,734		204,934		224,522		178,397	
Unique teachers	83,454		87,635		85,530		79,560	
Schools	4,999		4,834		4,803		4,446	

Note: The table reports means and standard deviations for the DiD analysis sample. The sample includes teachers under the age of 50 employed in public and subsidized schools between 2012 and 2018. Pre-reform refers to 2012–2015, post-reform to 2016–2018. SIMCE scores and parental background variables are available for approximately 88% of teacher-year observations (93% in public and 83% in subsidized schools).

Table B4: Summary statistics for the TVA subsample

	All (2013–2014)		TVA sample	
	Mean	SD	Mean	SD
Teacher characteristics				
Female	0.76	0.43	0.73	0.44
Age	35.0	7.3	35.2	7.5
Experience (years)	7.8	7.1	7.8	6.9
Tenure at current school	4.6	5.4	4.7	5.3
University graduate	0.89	0.31	0.95	0.22
Degree duration (semesters)	8.8	1.4	8.9	1.3
Contracted hours (weekly)	35.1	9.1	36.9	6.9
Primary school teacher	0.54	0.50	0.68	0.47
Exit rate (5-year)	0.039	0.193	0.018	0.134
Share in 2013	0.49	0.50	0.28	0.45
Share in 2014	0.51	0.50	0.72	0.45
Rural	0.13	0.33	0.16	0.37
Public school	0.50	0.50	0.50	0.50
School characteristics				
Far North	0.06	0.24	0.06	0.23
North	0.07	0.26	0.07	0.25
Center	0.57	0.50	0.57	0.49
South	0.22	0.41	0.22	0.42
Far South	0.08	0.27	0.08	0.28
SIMCE math score	251.0	29.1	250.7	29.6
SIMCE reading score	247.9	22.7	247.0	22.7
Share mothers with college	0.09	0.11	0.09	0.11
TVA coverage				
Share with math TVA			0.47	
Share with reading TVA			0.53	
Observations	223,426		19,139	
Teachers	129,552		19,139	

Note: The table compares the full DiD sample to the subsample of teachers with pre-treatment teacher value-added (TVA) estimates. The *All* column includes all public and subsidized school teachers under 50 observed in 2013–2014. The *TVA sample* column shows teachers with at least one TVA measure (math or reading) in 2013–2014, measured at their most recent TVA year. TVA is estimated from a value-added model using student SIMCE test scores, controlling for lagged scores, classroom and school characteristics, with teacher, school type, and grade-by-year fixed effects.

Table B5: Relationship between teacher value-added and evaluation scores

	Before reform			After reform		
	Coeff.	SE	N	Coeff.	SE	N
Portfolio	0.033***	(0.004)	10,153	0.039***	(0.004)	9,550
AVDI / ECEP	0.023***	(0.005)	5,265	0.025***	(0.003)	9,207

Note: Each cell reports the coefficient from a bivariate regression of teacher value-added (TVA) on the standardized evaluation score. TVA is the average of math and reading value-added estimates. Each evaluation observation is matched to the closest TVA measured within three years. The second evaluation instrument is AVDI before the reform (2012–2015) and ECEP after the reform (2016–2018). All evaluation scores are standardized within year. Robust standard errors in parentheses.

Table B6: Characteristics of low- and high-TVA teachers by sector

	Public		Subsidized	
	Low TVA	High TVA	Low TVA	High TVA
Teacher characteristics				
Female	0.723	0.743	0.730	0.746
Age	35.7	36.0	34.1	34.5
Experience (years)	7.9	8.5	7.1	7.9
Tenure at school (years)	4.4	5.1	4.3	5.2
Contracted hours (weekly)	36.6	37.5	36.6	37.8
University degree	0.942	0.951	0.950	0.956
School characteristics				
Rural school	0.207	0.307	0.071	0.052
Teaches primary	0.677	0.797	0.591	0.627
Teaches secondary	0.236	0.155	0.346	0.340
School SIMCE math	228.3	247.0	252.8	274.6
School SIMCE reading	229.2	244.8	246.1	262.7
Share mothers with college	0.034	0.042	0.132	0.162
Share students income Q1	0.388	0.407	0.167	0.138
Share students income Q5	0.036	0.041	0.183	0.228
Observations	4,474	4,468	4,395	4,388

Note: Means in 2014 for teachers with pre-reform value-added measured in 2013 or 2014. Teachers are classified as high or low TVA using subject-specific within-sector medians. Value added is estimated separately for mathematics and reading, and 5.0% of teachers have it in both subjects. For them the mathematics estimate is used.

Table B7: Career stage assignment based on performance in evaluation tests

Portfolio	**ECEP**			
	< 1.88	$[1.88, 2.75)$	$[2.75, 3.38)$	≥ 3.38
≥ 3.00	Advanced	Expert I	Expert II	Expert II
$[2.50, 3.00)$	Advanced	Advanced	Expert I	Expert II
$[2.25, 2.50)$	Early	Early	Advanced	Expert I
$[2.00, 2.25)$	Early	Early	Early	Advanced
< 2.00	Initial	Initial	Initial	Initial

Note: Career stage assignment as a function of Portfolio and ECEP score ranges. Eligibility for each stage also depends on a minimum number of years of experience: 4 years for Early and Advanced, 8 years for Expert I, and 12 years for Expert II.

Table B8: Summary statistics for evaluated teachers and all public school teachers

	All public		Exp ≥ 4		Evaluated	
	Mean	SD	Mean	SD	Mean	SD
2016						
Teacher characteristics						
Female	0.74	0.44	0.75	0.43	0.77	0.42
Age	35.0	7.1	37.5	6.4	37.7	6.4
Experience (years)	7.5	6.4	10.6	5.7	10.7	5.7
Contracted hours (weekly)	36.7	8.8	37.8	7.8	37.5	7.4
Exit rate (5-year)	0.038	0.191	0.022	0.147	0.017	0.128
School characteristics						
Rural	0.21	0.41	0.21	0.41	0.23	0.42
SIMCE math score	237.8	25.9	238.1	25.7	238.8	25.7
SIMCE reading score	240.2	21.4	240.7	21.3	241.8	21.2
Share mothers with college	0.05	0.06	0.05	0.06	0.05	0.06
Evaluation scores						
Portfolio score					2.53	0.27
ECEP score					2.65	0.51
Observations	64,705		42,470		9,129	
2017						
Teacher characteristics						
Female	0.73	0.44	0.75	0.43	0.75	0.43
Age	35.1	6.9	37.5	6.2	37.4	6.1
Experience (years)	7.6	6.3	10.4	5.6	10.4	5.4
Contracted hours (weekly)	37.7	8.5	38.8	7.4	39.2	6.5
Exit rate (5-year)	0.032	0.177	0.017	0.128	0.016	0.126
School characteristics						
Rural	0.21	0.41	0.21	0.41	0.21	0.41
SIMCE math score	240.3	23.9	240.8	23.8	239.1	23.9
SIMCE reading score	241.2	20.8	241.6	20.7	240.4	20.8
Share mothers with college	0.05	0.06	0.05	0.06	0.05	0.06
Evaluation scores						
Portfolio score					2.55	0.29
ECEP score					2.70	0.55
Observations	68,552		46,618		10,914	

Note: Each panel shows statistics for a given evaluation year. "All public" includes all public school teacher-year observations. "Exp ≥ 4" restricts to teachers with at least four years of experience. "Evaluated" further restricts to teachers evaluated in that year. Exit rate (5-year) in the first two columns equals one if the teacher is not observed in years $t+1$ through $t+5$. In the evaluated sample, exit equals one if this condition holds at the test year *or* the year after (stage assignment year). Evaluation scores are reported for teachers with a non-missing score.

Table B9: Covariate balance at the Early-to-Advanced threshold

	Mean	RD Estimate	Bandwidth	Eff. N
Teacher characteristics				
Female	0.761	-0.0006	0.380	9,724
		(0.0181)		
Age	37.4	-0.1576	0.324	8,725
		(0.2888)		
Experience (years)	10.6	0.2446	0.369	9,622
		(0.2380)		
Tenure at school	6.2	0.3912	0.286	7,779
		(0.2392)		
University graduate	0.888	0.0143	0.379	9,658
		(0.0139)		
Degree duration (sem.)	8.7	0.0912	0.276	7,507
		(0.0721)		
Primary education	0.609	-0.0255	0.272	7,562
		(0.0232)		
School characteristics				
Rural	0.209	0.0378	0.234	6,546
		(0.0231)		
Center	0.488	-0.0169	0.284	7,779
		(0.0241)		
South	0.272	0.0265	0.322	8,725
		(0.0202)		
SIMCE math	239.4	0.1900	0.297	7,691
		(1.0719)		
SIMCE reading	241.2	0.3436	0.254	7,017
		(1.0468)		
Share mothers college	0.050	0.0065**	0.263	7,149
		(0.0028)		

Note: Each row reports local polynomial RD estimates of the discontinuity in a pre-determined covariate at the Early-to-Advanced Portfolio threshold, following Calonico et al. (2019). The running variable is the recentered Portfolio score. Bandwidth selected by MSE-optimal procedure for each covariate. Triangular kernel. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B10: Effect of reaching the Advanced stage on teacher exit

	Five-year exit		One-year exit	
	Conventional	Robust	Conventional	Robust
No controls	0.0002	0.0012	-0.0056	-0.0055
	(0.0049)	(0.0057)	(0.0080)	(0.0096)
ECEP category fixed effects	-0.0000	0.0010	-0.0062	-0.0060
	(0.0048)	(0.0056)	(0.0081)	(0.0096)
ECEP category and test year fixed effects	-0.0003	0.0007	-0.0065	-0.0066
	(0.0049)	(0.0057)	(0.0081)	(0.0096)
Covariates	0.0020	0.0030	-0.0031	-0.0026
	(0.0048)	(0.0056)	(0.0078)	(0.0092)
Fixed effects and covariates	0.0016	0.0027	-0.0037	-0.0030
	(0.0048)	(0.0055)	(0.0079)	(0.0094)

Note: Local linear RD estimates following Calonico et al. (2019), with a triangular kernel and an MSE-optimal bandwidth selected separately for each estimate. The running variable is the recentered Portfolio score at the Early-to-Advanced threshold. Conventional estimates use the standard local linear estimator, and robust estimates are bias-corrected with robust standard errors. Standard errors in parentheses, adjusted for mass points in the running variable. Covariates are gender, age, experience, contracted hours, tenure at school, rural school, region, and the school SIMCE scores in mathematics and reading and share of mothers with college education.

C Derivations for the conceptual framework

This appendix provides formal derivations for the results in Section 3.

C.1 Setup and notation

Teachers have ability a, concave utility $u(\cdot)$ over income ($u' > 0$, $u'' < 0$), and face an outside option $\omega(a)$. The wage is w_0 and $\Delta > 0$ is a uniform increase in it. Teaching provides non-pecuniary benefit $\bar{\mu}(a) + \varepsilon$, with ε logistic with scale σ.

The exit probability at wage w is

$$h(a, w) = \Lambda\left(\frac{u(\omega(a)) - u(w) - \bar{\mu}(a)}{\sigma}\right) \equiv \Lambda(z(a, w)),$$

where $z(a, w) = [u(\omega(a)) - u(w) - \bar{\mu}(a)]/\sigma$ is the net outside value. Teachers with higher z are more likely to exit.

How exit responds to a wage increase. Raising the wage from w to $w + x$ changes exit by

$$h(a, w + x) - h(a, w) \;=\; \Lambda\big(z(a, w) - g\big) - \Lambda\big(z(a, w)\big), \qquad g \equiv \frac{u(w + x) - u(w)}{\sigma} > 0, \tag{5}$$

which for small g is approximately $-\lambda(z(a, w)) \cdot g$. The size of the reduction depends on $\lambda(z)$, which measures how close the teacher is to the exit margin, and on g, the utility the increase buys. The logistic density λ is symmetric and single-peaked at $z = 0$ and increasing on $(-\infty, 0]$. I restrict attention to $z < 0$, that is, to exit probabilities below one half. In that region teachers with higher baseline exit rates are closer to the exit margin, and any increase in the wage moves a teacher farther from it.

The bonus. On top of Δ, an amount B is paid to teachers whose evaluation score $s = a + \nu$ crosses a threshold τ, which happens with probability $p(a) = 1 - F_\nu(\tau - a)$, increasing in a. The payment is binary, so expected exit at the new wages is the mixture

$$\big(1 - p(a)\big)\, h\big(a,\, w_0 + \Delta\big) \;+\; p(a)\, h\big(a,\, w_0 + \Delta + B\big). \tag{6}$$

C.2 A wage increase reduces exit

The treatment effect for a teacher with ability a is the difference between expected exit at the new wages, given in equation (6), and exit at w_0,

$$TE(a) = \big(1 - p(a)\big)\, h\big(a,\, w_0 + \Delta\big) + p(a)\, h\big(a,\, w_0 + \Delta + B\big) - h(a,\, w_0).$$

Both new wages exceed w_0, since $\Delta > 0$ and $B \geq 0$. By strict monotonicity of u this implies $z(a, w_0 + \Delta) < z(a, w_0)$ and $z(a, w_0 + \Delta + B) \leq z(a, w_0 + \Delta)$, and since Λ is strictly increasing, $h(a, w_0 + \Delta + B) \leq h(a, w_0 + \Delta) < h(a, w_0)$. A weighted average of the two is therefore below $h(a, w_0)$, so $TE(a) < 0$ for every a, whether or not the teacher receives B. □

C.3 Attenuation of the bonus

The discontinuity at τ compares teachers just above and below the threshold, both of whom have already received Δ,

$$\delta_\Delta \equiv h(a,\, w_0 + \Delta + B) - h(a,\, w_0 + \Delta).$$

The benchmark is the same payment made from w_0, that is, $\delta_0 \equiv h(a, w_0 + B) - h(a, w_0)$. Applying equation (5) to each, and writing $z_\Delta = z(a, w_0 + \Delta)$ and $z_0 = z(a, w_0)$, the two contrasts are

$$\Lambda(z_\Delta) - \Lambda(z_\Delta - g_\Delta) \qquad \text{and} \qquad \Lambda(z_0) - \Lambda(z_0 - g_0),$$

with $g_\Delta = [u(w_0 + \Delta + B) - u(w_0 + \Delta)]/\sigma$ and $g_0 = [u(w_0 + B) - u(w_0)]/\sigma$. Two separate forces make the first smaller than the second.

Distance from the exit margin. The increase Δ lowers the net outside value, $z_\Delta < z_0$, moving the teacher farther from the exit margin. Write the reduction in exit probability as $\int_{z-g}^{z} \lambda(t)\, dt$. The logistic density λ is increasing on $(-\infty, 0]$, so if $z_0 \leq 0$ the same interval of length g integrates to less when shifted left. Holding g fixed, B therefore produces a smaller reduction in exit probability when paid on top of Δ than when paid from w_0. The condition $z_0 \leq 0$ is that the baseline exit rate be at most one half.

Size of the utility gain. By strict concavity of u, for any $B > 0$ and $\Delta > 0$,

$$u(w_0 + \Delta + B) - u(w_0 + \Delta) < u(w_0 + B) - u(w_0),$$

since u' is strictly decreasing. Hence $g_\Delta < g_0$, and the exit reduction is increasing in g. The larger is Δ relative to B, the smaller the utility B buys. □

C.4 Derivation of the ability gradient

Ability enters the treatment effect through two distinct objects. It shifts the teacher's distance from the exit margin, $z(a)$, and it shifts the probability of receiving B, namely $p(a)$. Therefore,

$$\frac{\partial\, TE}{\partial a} = \underbrace{\Big[\lambda\big(z(a) - D\big) - \lambda\big(z(a)\big)\Big] \cdot z'(a)}_{\text{distance from the margin}} + \underbrace{p'(a) \cdot \Big[h\big(a, w_0 + \Delta + B\big) - h\big(a, w_0 + \Delta\big)\Big]}_{\text{bonus exposure}} + O(p), \tag{7}$$

where $z(a) \equiv z(a, w_0)$, $D \equiv [u(w_0 + \Delta) - u(w_0)]/\sigma > 0$ is the normalized utility gain from Δ, and the remainder collects the terms of order p.

Distance from the exit margin. Since $D > 0$ we have $z(a) - D < z(a)$, and the logistic density $\lambda(z) = \Lambda(z)(1 - \Lambda(z))$ is symmetric and single-peaked at $z = 0$. If $z(a) < 0$, both arguments are negative and $z(a) - D$ is farther from zero, so $\lambda(z(a) - D) < \lambda(z(a))$ and the bracket is negative. A baseline exit rate below one half means exactly $z(a) < 0$, so the bracket is negative.

The sign of the gradient then depends on

$$z'(a) = \frac{u'(\omega(a))\, \omega'(a) - \bar{\mu}'(a)}{\sigma}.$$

Better outside options for more able teachers, $\omega'(a) > 0$, push z up and toward exit. Greater intrinsic satisfaction for more effective teachers, $\bar{\mu}'(a) > 0$, pushes z down and toward staying.

Case (a), outside option dominates. If $u'(\omega)\, \omega' > \bar{\mu}'$ then $z'(a) > 0$, so high-ability teachers have higher baseline exit rates. Since the bracket is negative and $z'(a) > 0$, the treatment effect is more negative for high-ability teachers, so they are retained more.

Case (b), non-pecuniary channel dominates. If $\bar{\mu}' > u'(\omega)\,\omega'$ then $z'(a) < 0$, so low-ability teachers have higher baseline exit rates. The treatment effect is more negative for them, so they are retained more. □

Role of imperfect measurement. The correlation between a and the score $s = a + \nu$ determines how well B targets high-ability teachers. When ν has large variance, $p(a)$ varies little across ability levels. In the limit of pure noise, $\text{Var}(\nu) \to \infty$ and $p(a) \to 1/2$ for all a, so B becomes a flat payment and the bonus exposure term vanishes.

D Imputed wages

Administrative records from MINEDUC do not include teacher salaries directly. I therefore construct imputed wages for each teacher-year observation by applying the statutory formulas from the *Estatuto Docente* (DFL 1, 1996, as modified by Law 20,903) to observable teacher characteristics. The salary essentially depends on four variables: teaching level, contracted hours, years of experience, and career stage.

Base wage. The base component is the *Remuneración Básica Mínima Nacional* (RBMN), set by the *Escala Única de Remuneraciones*. The RBMN varies by teaching level. For 2015, the hourly rates for a 44-hour contract were: \$12,293 for *educación básica*, \$12,935 for *educación media*, and \$17,845 for *educación parvularia*. In 2015, 65.6% of teachers in the sample are classified as *básica*, 26.7% as *media*, and 7.7% as *parvularia*. The base wage for teacher i in year t is:

$$W_{it}^{base} = r_{\ell(i)} \times h_{it},$$

where $r_{\ell(i)}$ is the hourly RBMN for the teacher's level ℓ and h_{it} denotes contracted weekly hours.

Experience component. The bonus related to experience is a percentage of the base wage that increases with biennia of service, capped at 15 biennia (30 years). Before the reform (Article 48 of the *Estatuto Docente*), the rates at which experience was increasing the salary were 6.76% for the first biennium and 6.66% for each additional biennium. The reform halved these rates to 3.38% and 3.33%, respectively. Define $b_{it} = \min\big(\lfloor \text{experience}_{it}/2 \rfloor,\, 15\big)$. The experience bonus is:

$$W_{it}^{exp} = \begin{cases} \big(0.0676 + 0.0666 \times \max(b_{it} - 1,\, 0)\big) \times W_{it}^{base} & \text{if } t \leq 2016, \\ \big(0.0338 + 0.0333 \times \max(b_{it} - 1,\, 0)\big) \times W_{it}^{base} & \text{if } t \geq 2017, \end{cases}$$

with $W_{it}^{exp} = 0$ when $b_{it} = 0$.

Career stage bonus. From 2017 onward, teachers assigned to a career stage receive an additional bonus (*Asignación por Tramo de Desarrollo Profesional*, Article 49). This bonus has two components. The first, which is variable, depends on both contracted hours (relative to a 44-hour contract) and biennia of experience (relative to the 15-biennia maximum). The second, a fixed component (in the sense that it does not depend on

experience), is paid only to teachers in the Advanced stage or above and depends only on contracted hours. The maximum values of both components are reported in Table B1. The total career stage bonus for teacher i is

$$W_{it}^{tramo} = V_s \times \frac{h_{it}}{44} \times \frac{b_{it}}{15} + F_s \times \frac{h_{it}}{44},$$

where V_s and F_s denote the variable and fixed amounts for stage s.

Professional recognition bonus. The *Bonificación de Reconocimiento Profesional* (BRP, Article 54) is paid to teachers holding a professional degree from a university, *instituto profesional* or *escuela normal*. It is fixed and independent of experience and career stage. Roughly 96.8% of teachers in the sample are eligible. The reform raised it from \$61,538 to \$224,861 per month for a 30-hour contract.

$$W_{it}^{brp} = \begin{cases} 61{,}538 \times \mathbf{1}[\text{eligible}_i] & \text{if } t \leq 2016, \\ 224{,}861 \times \mathbf{1}[\text{eligible}_i] & \text{if } t \geq 2017. \end{cases}$$

Because eligibility is nearly universal, the BRP contributes almost nothing to cross-sectional variation, but it accounts for most of the change in wage levels between 2016 and 2017.

Total imputed wage. The imputed monthly wage is:

$$W_{it} = \begin{cases} W_{it}^{base} + W_{it}^{exp} + W_{it}^{brp} & \text{if } t \leq 2016, \\ W_{it}^{base} + W_{it}^{exp} + W_{it}^{tramo} + W_{it}^{brp} & \text{if } t \geq 2017. \end{cases}$$

E Placebo test with centralized admissions

Chile's Ley de Inclusión (Law 20,845) replaced school-level admissions in subsidized schools with the Sistema de Admisión Escolar (SAE), a centralized lottery. The SAE was rolled out regionally over several years, as detailed in Table E11. I exploit this staggered rollout to test whether the SAE itself affected teacher exit rates.

Table E11: SAE regional rollout

SAE year	Regions	Share of schools (%)
2017	Tarapacá, Coquimbo, O'Higgins, Los Lagos	23.2
2018	Arica y Parinacota, Antofagasta, Atacama, Valparaíso, Maule, Biobío, Araucanía, Los Ríos, Aysén	49.7
2019	Metropolitana	27.2

Note: Share of schools computed from the analysis sample in 2015. Magallanes is excluded (2016 pilot with 63 schools). Ñuble is excluded (created in 2018, no pre-period).

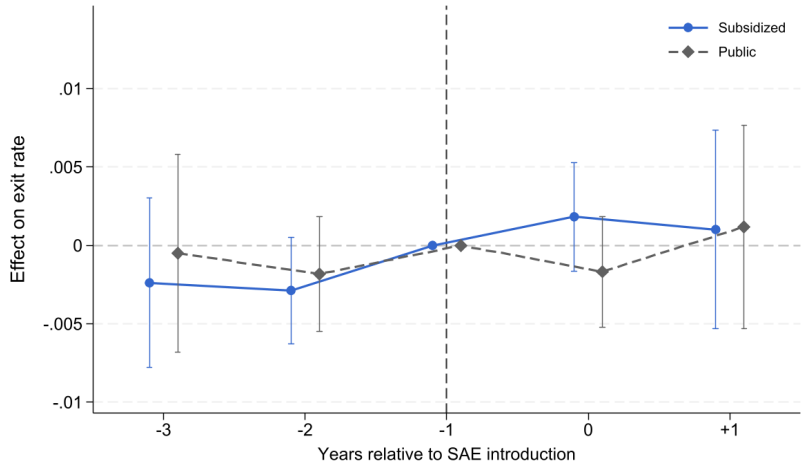
The specification is a staggered difference-in-differences event study. Let $r(i)$ denote the region of school i, and let g_r denote the SAE adoption year for region r. I define relative time as $k_{it} = t - g_{r(i)}$ and estimate:

$$\text{Exit}_{it} = \sum_{k \neq -1} \beta_k \cdot \mathbf{1}[k_{it} = k] + \alpha_i + \delta_t + \varepsilon_{it}, \tag{8}$$

where α_i are school fixed effects, δ_t are year fixed effects, and the sum runs over $k \in \{\leq -5, -4, -3, -2, 0, +1\}$, with $k = -1$ as the omitted baseline and all leads at $k \leq -5$ binned into a single indicator. Standard errors are clustered at the school level. I estimate the model separately for subsidized and public school teachers.

Results are shown in Figure E14.

Figure E14: SAE rollout and teacher exit



Note: The figure reports event study coefficients β_k from equation (8), estimated separately for subsidized and public school teachers. The baseline is $k = -1$. Leads at $k \leq -5$ and $k = -4$ are included in the regression but not plotted. 95% confidence intervals shown.

F Wage spillovers

Data. I use data from the Chilean national socioeconomic household survey (CASEN) for 2013 and 2017. The CASEN is a repeated cross-section conducted every two years by the Ministry of Social Development, with sample sizes of approximately 200,000 to 270,000 individuals per wave.

The wage measure is the corrected total labor income constructed by the CASEN team, which aggregates salary, overtime, bonuses, and secondary job income with imputation for non-response. This variable is consistently defined across both waves. I deflate wages to 2017 pesos using the annual average consumer price index.

Population. I identify teachers using four-digit occupation codes from the CIUO-88 classification (secondary teachers, primary and preschool teachers at higher and mid-level qualifications) combined with four-digit economic activity codes from the CIIU Rev. 3 classification (primary education, general secondary education, and technical secondary education). I classify teachers as public or private based on the employment category variable, which distinguishes public sector employees (central government or municipal) from private sector employees. This variable does not separate subsidized from fully private schools. However, as discussed before, private schools are only a small share of non-public schools.

Summary statistics. Table F12 reports summary statistics for the teacher sample by sector and year. The two groups have similar gender composition and years of schooling in both years. Public teachers are on average about two years older. In 2013 they also worked about four hours less per week than private sector teachers, while in 2017 the two groups worked similar hours. In 2013, public teachers earned substantially less than private sector teachers (534,000 vs. 608,000 CLP).

Comparing wages over time. To account for compositional differences across survey waves, I residualize wages on a rich set of observable characteristics: a second-order polynomial in age, gender, years of schooling, weekly hours worked, region fixed effects, an urban indicator, four-digit occupation code fixed effects, and marital status. I then add back to the residuals the unconditional mean for interpretability. Table F13 reports the results. Before the reform, the ratio of public to private composition-adjusted wages was 0.87. By 2017, public wages had grown by 18.4% in real terms, closing the

Table F12: Summary statistics for the CASEN teacher sample

	2013		2017	
	Public	Private	Public	Private
Age	40.8	38.4	40.8	39.2
Female (%)	77.6	79.0	77.0	75.4
Years of schooling	16.4	16.4	16.3	16.5
Weekly hours	41.2	45.3	40.5	39.9
Married (%)	45.6	41.6	36.3	39.7
Monthly wage (CLP)	534,416	607,725	689,757	680,363
Observations	1,252	1,097	1,317	1,295

Note: Weighted means using CASEN survey weights. Monthly wage is the corrected total labor income in nominal CLP. Teachers are identified by CIUO-88 occupation codes and CIIU Rev. 3 activity codes for the education sector. Public/private classification is based on the self-reported employment category.

gap entirely. Private teacher wages, by contrast, were essentially unchanged (-0.1%), consistent with the absence of a competitive wage response from subsidized schools.

Table F13: Composition-adjusted real teacher wages by sector (2017 CLP)

	2013	2017	Change
Mean residualized real wage			
Public	591,231	699,744	18.4%
Private	676,125	675,124	-0.1%
Ratio (Public/Private)	0.87	1.04	
Observations			
Public	1,245	1,300	
Private	1,091	1,287	

Note: Wages are corrected total labor income from the CASEN, deflated to 2017 pesos using the annual average CPI. Residualized wages are obtained by regressing real wages on age, age squared, gender, years of schooling, weekly hours worked, region fixed effects, urban/rural, occupation code fixed effects, and marital status, using survey weights, and adding back the grand mean. Teachers are identified by CIUO-88 occupation codes and CIIU Rev. 3 activity codes for the education sector. Public/private classification is based on the self-reported employment category.